

MODE COLLAPSE UNDER GRPO: MODEBENCH AND X-MODE GRPO

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ABSTRACT

Repeated sampling helps only if a policy proposes distinct viable solutions. Binary-reward RL can raise accuracy while *erasing* breadth: equally rewarded correct answers compete through sampling frequency, so eight attempts become copies. We introduce MODEBENCH, six executable domains where execution determines correctness and canonical identity: five static-response and one interactive MuJoCo maze. We introduce x-Mode GRPO, which adds verified replay to Dr.GRPO: one validator-positive exemplar per observed execution mode is retained and revisited with uniform likelihood. In a finite categorical mean flow, any positive recurrent dose prevents banked modes from vanishing; full verified coverage rules out winner-take-all collapse. A dated interim snapshot is positive in every currently paired breadth panel, but coverage is incomplete and no final performance claim is made. Accuracy conceals collapse; MODEBENCH makes monotony countable and x-Mode GRPO makes retention testable.

1 INTRODUCTION

Inference-time scaling promises to trade inference compute for better reasoning by drawing multiple candidates and selecting among them. We study a failure that breaks that trade: binary-verifier Reinforcement Learning (RL) Language Model (LM) post-training can raise accuracy while concentrating a policy on one successful execution. Under group-relative binary-reward RL, eight nominal attempts can become eight copies of the same answer. As we show in Figure 1, standard Group Relative Policy Optimization (GRPO) [24, 20] looks better by correctness after training while its useful proposal support quietly collapses.

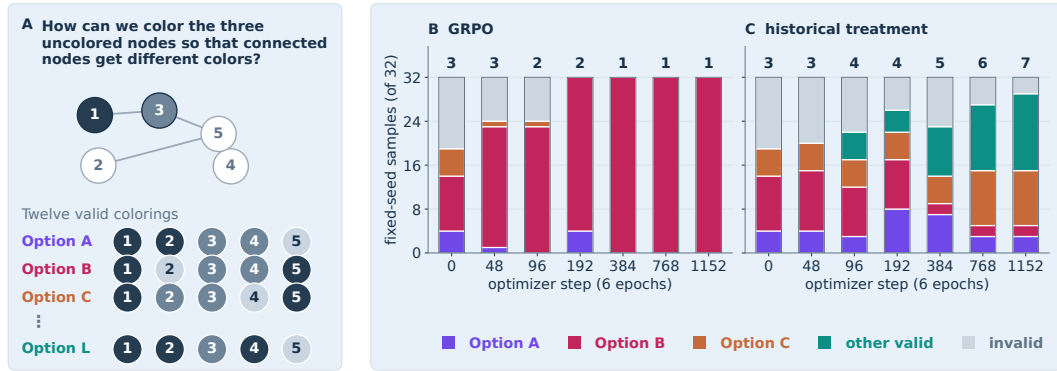


Figure 1: **Mode-collapse example.** (A) One prompt has 12 valid colorings; three of them and the twelfth are shown. (B–C) Four fixed-seed $K = 8$ draws from matched Qwen2.5-0.5B-Instruct trajectories; color denotes an executed correct mode and gray an invalid response. Dr.GRPO reaches 32/32 correct but contracts to one mode; a historical multi-component treatment retains seven. This panel motivates the retention question but is not evidence for x-Mode GRPO. Accuracy rises while the marginal value of sampling collapses.

Inference-time scaling requires genuine solution diversity. Self-consistency [28], best-of- N [2], verifier-guided search [26], and self-improvement [6] all generate candidates, and a selector returns only a route the policy proposes. Breadth therefore decides whether more inference buys another

chance or another copy: RLVR can win at small k yet lose at large k [31], while RLHF and fine-tuning can trade generalization or $\text{pass}@K$ for narrower outputs [12, 5].

Useful response breadth is difficult to define and preserve. Raw strings overcount formatting aliases, while final answers can merge distinct executable programs or procedures. In exact tasks, validator-produced canonical execution keys provide task-relevant identity. Valid modes are unknown, but a validator-positive execution can still be retained after it is observed, and rare modes can vanish before a prompt returns [23]. Binary reward equates all correct responses [24, 20], while finite on-policy sampling reinforces common modes and leaves absent modes untouched [4]. Repairing the failure therefore takes a signal bound to executed mode identity.

Prior work documents collapse and promotes outcome bonuses, batch penalties, $\text{pass}@K$ objectives, and set-level or distributional optimization [4, 5, 27, 29, 3, 16]. Existing work nevertheless lacks an execution-grounded benchmark in which the same validator determines correctness and outcome identity. x-Mode GRPO adds one intervention to Dr.GRPO: verified replay retains a validator-positive output beyond its collection batch.

We introduce MODEBENCH, whose six domains derive correctness and canonical mode identity from the same execution, and evaluate x-Mode GRPO over the policy’s own accepted outcomes. Five are static-response; PointMaze asks the same Dr.GRPO-versus-x-Mode GRPO question of an LM acting sequentially in an interactive MuJoCo environment. The registered evaluation spans aligned Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B comparisons. Throughout, *x-Mode GRPO* names only Dr.GRPO with verified replay as its sole auxiliary derivative; *verified replay* names that mechanism. Older multi-component treatments are labeled historical and never attributed to

Summary of contributions.

- *Problem and measurement* (Sections 2–3): MODEBENCH separates correctness from executed correct-mode support across five static domains and one interactive domain.
- *Mechanism and analysis* (Section 4 and Appendix A): x-Mode GRPO uses verified replay to preserve discovered correct modes; categorical analysis proves banked-mode non-extinction and excludes winner-take-all collapse after verified coverage.
- *Evidence* (Sections 6–7): five-seed Dr.GRPO trajectories document correct-mode collapse; clean Dr.GRPO-versus-x-Mode GRPO evaluation spans six domains and three model scales.

2 MODE COLLAPSE UNDER GRPO

The unit of diversity must be task-relevant and execution-grounded. Surface strings overcount formatting changes, while final answers can erase different programs or procedures. We therefore identify a mode with a validator-positive executed outcome: the same execution determines whether a response is correct and which canonical outcome it realizes. This section fixes notation and formalizes the failure the contract exposes.

2.1 PROMPTS, VALIDATORS, AND POLICY-INDUCED QUANTITIES

Let $\mathcal{X}$ denote prompts and $\mathcal{Y}$ responses. Every prompt $x \in \mathcal{X}$ carries a reference specification s_x and a finite nonempty set $\mathcal{C}_x$ of *correct execution modes* (canonical keys). A *validator* is a map

$$V(\cdot, s_x) : \mathcal{Y} \rightarrow \mathcal{C}_x \cup \{\perp\}$$

that executes a response and returns either the canonical key of its verified outcome or the failure symbol $\perp$; the binary task reward is $r(x, y) = \mathbf{1}[V(y, s_x) \neq \perp]$. The learner observes only V ’s output on the responses it produces: neither $\mathcal{C}_x$ nor $|\mathcal{C}_x|$ is available to it.

A policy $\pi_\theta(\cdot | x)$ over $\mathcal{Y}$ induces two objects. The *correct mass* $P_\theta^+(x) = \Pr_{Y \sim \pi_\theta(\cdot | x)}[V(Y, s_x) \neq \perp]$ is the probability that a sample verifies. Whenever $P_\theta^+(x) > 0$, the *conditional correct-mode distribution* on $\mathcal{C}_x$ is

$$p_\theta^+(c | x) = \Pr_{Y \sim \pi_\theta(\cdot | x)}[V(Y, s_x) = c \mid V(Y, s_x) \neq \perp].$$

Two population summaries of p_θ^+ measure breadth: the verified-mode entropy $H_\theta^+(x) = -\sum_{c \in \mathcal{C}_x} p_\theta^+(c | x) \log p_\theta^+(c | x)$ and the concentration $\max_{c \in \mathcal{C}_x} p_\theta^+(c | x)$. These are diagnostic population quantities, not training inputs.

Definition 2.1 (Mode collapse). *A training trajectory $(\theta_t)_{t \geq 0}$ exhibits mode collapse on x when $p_{\theta_t}^+(\cdot | x)$ concentrates on a strict subset of $\mathcal{C}_x$ as t grows while $P_{\theta_t}^+(x)$ remains comparable or increases.*

Mode collapse is therefore a statement about where correct mass sits rather than accuracy: a high-accuracy policy concentrated on one correct mode exhibits mode collapse.

2.2 SAMPLE DIAGNOSTICS

H_θ^+ and the concentration are unavailable when the support is unknown, so evaluation estimates breadth from samples. Fix $K \in \mathbb{N}$ and draw $Y_1, \dots, Y_K \stackrel{\text{iid}}{\sim} \pi_\theta(\cdot | x)$ at temperature one. Define

$$D_K(x) = |\{V(Y_i, s_x) : i \leq K\} \setminus \{\perp\}|, \quad S_K(x) = \mathbf{1}[D_K(x) > 0], \quad F_K(x) = \frac{1}{K} \sum_{i=1}^K r(x, Y_i),$$

the number of distinct verified modes among the K draws, the indicator that at least one draw verifies, and the fraction of draws that verify. With K draws jointly, evaluation reports

$$\text{distinct@}K = \mathbb{E}[D_K], \quad \text{pass@}K = \mathbb{E}[S_K], \quad \text{mean@}K = \mathbb{E}[F_K],$$

together with greedy $\text{pass@}1 = \mathbb{E}_x[r(x, y^{\text{gr}}(x))]$ for $y^{\text{gr}}(x)$ the deterministic greedy decode. $\text{distinct@}K$ counts correct execution modes; $\text{pass@}K$ asks whether search finds any. The three joint-sample metrics form an ordered diagnostic; greedy $\text{pass@}1$ is reported separately.

Lemma 2.2 (Ordered sample diagnostics). *(i) $F_K \leq S_K \leq D_K$ pointwise, hence $\text{mean@}K \leq \text{pass@}K \leq \text{distinct@}K$, and the first inequality is strict in expectation unless, almost surely, either no draw or every draw verifies. (ii) Because $D_K \mathbf{1}[D_K > 0] = D_K$,*

$$\frac{\text{distinct@}K}{\text{pass@}K} = \mathbb{E}[D_K | D_K > 0] =: \text{modes-per-success},$$

the mean number of different correct executions a successful K -sample set contains; a value of 1 means each successful set contains only one distinct execution. That execution may differ across prompts or independent sample sets.

Proof. (i) $F_K \in [0, 1]$ with $F_K > 0 \Rightarrow S_K = 1$, and $S_K = 1 \Rightarrow D_K \geq 1$; moreover $F_K = S_K$ exactly when the number of verifying draws is 0 or K . (ii) $\mathbb{E}[D_K] / \Pr[D_K > 0] = \mathbb{E}[D_K \mathbf{1}[D_K > 0]] / \Pr[D_K > 0] = \mathbb{E}[D_K | D_K > 0]$. $\square$

A tight band among $\text{mean@}K$, $\text{pass@}K$, and $\text{distinct@}K$ is therefore a sample-level signature of collapse: successful K -sample sets reveal almost no additional execution beyond the first. Agreement with greedy $\text{pass@}1$ is a separate empirical observation rather than part of the ordering.

2.3 WHY EQUAL REWARD CONCENTRATES CORRECT MODES

Group-relative task advantages. Fix a prompt x and draw a group $Y_1, \dots, Y_G \stackrel{\text{iid}}{\sim} \pi_\theta(\cdot | x)$ with $G \geq 2$ and rewards $r_i = r(x, Y_i)$. Both variants center the binary reward on its group mean, $A_i^{\text{task}} = r_i - G^{-1} \sum_{j=1}^G r_j$; Dr.GRPO then applies one shared maximum-length normalization where GRPO divides by the group reward standard deviation. Every quantity in these rescalings is a function of $(r_1, \dots, r_G)$ alone, and r_i is blind to *which* key V returned, so correct responses in the same group receive the same task advantage. That symmetry equalizes credit across identities and leaves mode ratios free to follow sampling frequency; Lemma A.1 shows that the expected GRPO and Dr.GRPO updates are positive scalar multiples of ∇P , so they share a direction and differ only in speed.

A categorical toy model. Fix x and suppress it from the notation. Partition the response categories as $\mathcal{A} = \mathcal{C} \cup \mathcal{N}$, with $m = |\mathcal{C}| \geq 2$ correct modes and $n = |\mathcal{N}| \geq 1$ incorrect categories; give

each category $a \in \mathcal{A}$ a logit z_a and set $p_a = \text{softmax}(z)_a$, correct mass $P = \sum_{c \in \mathcal{C}} p_c$, and conditional mode distribution $q_c = p_c/P$ —the categorical projections of $P_\theta^+(x)$ and $p_\theta^+(\cdot | x)$, in the notation of Appendix A. It therefore suffices to inspect gradient ascent on the expected binary reward $\mathbb{E}_{A \sim p} \mathbf{1}[A \in \mathcal{C}] = P$. For $c, d \in \mathcal{C}$,

$$\dot{z}_c = \frac{\partial P}{\partial z_c} = p_c(1 - P), \quad \frac{d}{dt} \log \frac{q_c}{q_d} = \frac{d}{dt} \log \frac{p_c}{p_d} = (p_c - p_d)(1 - P).$$

While incorrect mass remains ($P < 1$), the more common of two equally rewarded correct modes receives the larger logit update—and the conditional ratios q_c/q_d inherit the same dynamics. Finite on-policy groups sharpen the asymmetry: an absent rare mode receives no update, and an all-correct group has zero centered task advantage. Thus an early overrepresented correct mode can become a self-reinforcing and eventually locally frozen solution. In this categorical toy model, the binary-reward dynamics amplify sampling-frequency differences between correct modes. Recent analyses reach related conclusions through outcome-frequency amplification, divergence choice, or set-level policy geometry [25, 17, 15].

Appendix A makes this precise. Theorem A.2 isolates the winner-take-all mechanism under infinitesimal on-policy mean dynamics. The positive result, Theorem A.4 with Corollary A.5, is deliberately post-discovery. Under the theorem’s fixed-bank categorical assumptions, recurrent verified replay prevents banked modes from vanishing; with full verified coverage, the flow converges to the uniform correct-mode distribution. It does not prove discovery or global convergence in a nonconvex language model.

3 MODEBENCH: EXECUTABLE MODE MEASUREMENT

3.1 BENCHMARK

A benchmark domain is mode-verifiable only when V satisfies two requirements: (1) it executes the object the model actually generated and judges that execution, (2) the canonical key is derived from the same execution that established correctness. Additionally, cosmetic aliases merge; different executed outcomes stay distinct. This restricts diversity credit to verified responses and confines breadth to solutions the validator has distinguished.

Each domain trains on 384 prompts and evaluates on a fixed 128-prompt split, and the mean number of correct execution modes per evaluation prompt—exact over the released multi-answer split—is 6.27 for Graph coloring, 4.52 for Countdown, 229.44 for Python factors, 5.00 for MathIR, and 18.19 for PantryPlan; each PointMaze map exposes three to five separated corridors. Table 3 gives each domain’s generated object, executed key, and merged aliases; the exact prompt each policy receives is reproduced verbatim in Appendix E.

Graph coloring. Each prompt gives a graph, a partial three-coloring, and hidden vertices; the validator reconstructs the completion, preserves shown colors, and rejects monochromatic edges. The color vector is the key.

Countdown. Each prompt gives three distinct integers and a target, each usable once with $+$, $-$, $\times$, $\div$ and parentheses. The parser executes in exact rational arithmetic and canonicalization sorts commutative children, merging syntactic aliases while retaining genuinely different trees.

Python factors. Each prompt asks for `lambda n: EXPR` on four composite inputs; a syntax-restricted external worker (Appendix D) requires every returned d to satisfy $1 < d < n$ and $n \bmod d = 0$. The output vector is the key, so different program texts with identical behavior merge. A prompt with inputs $(n_1, \dots, n_m)$ has exactly $\prod_j |D(n_j)|$ modes for $D(n)$ the proper nontrivial divisors; released rows contain 16–3,600.

MathIR. Each prompt shows a linear equation and at most eight lettered transformations, of which the model emits at most four. An interpreter applies each to both sides and normalizes exactly after every step; the key is the state trajectory. Each prompt has five exhaustively enumerated valid routes, but these reference routes are not supplied to the policy or used to initialize replay.

PantryPlan. Each prompt asks for an exact integer-gram allocation from six frozen ingredients subject to inventory, serving-step, mass, nutrient, and dietary-tag constraints. The same decimal-arithmetic validator accepts the allocation and keys it by the sorted set of ingredients used: quantity and ordering aliases merge, while different feasible formulations remain distinct. The registered

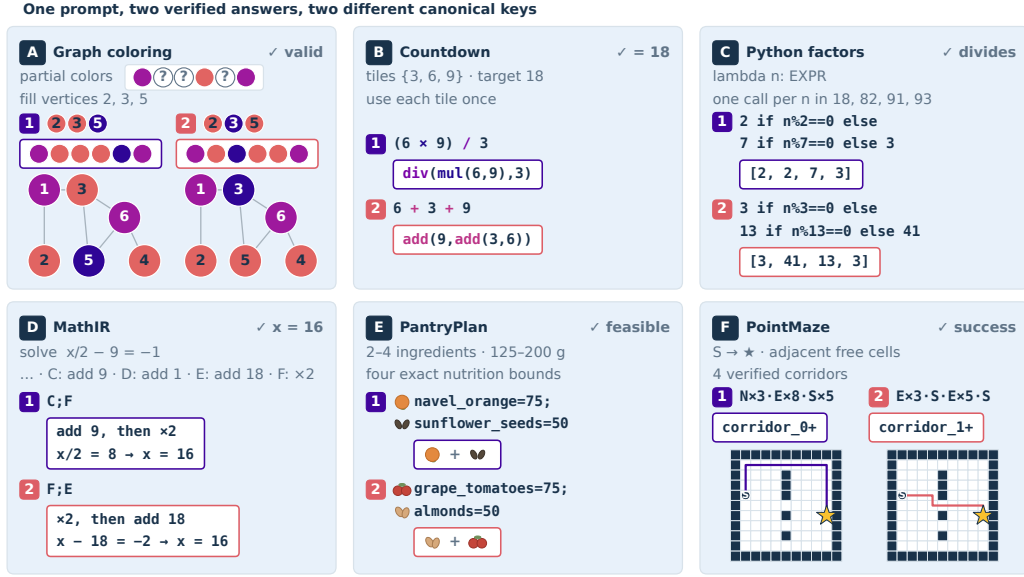


Figure 2: **Two verified modes across six executable environments.** Every answer or episode is paired with the canonical key produced by the same execution that accepted it. Graph answers complete $2??1?2$; Countdown keys are normalized ASTs; Python keys are executed return vectors; MathIR keys are state trajectories; PantryPlan keys are ingredient supports; and PointMaze keys are directed corridors crossed by the MuJoCo trajectory. Panel D prints the tail of that row’s six-action menu, so its leading ellipsis elides two actions and D is a legal action neither answer selects. In panel F, each plotted path joins the LM’s sequential adjacent-cell choices for display. The five static panels use their frozen evaluation splits; the PointMaze panel uses a simulator-certified interface-development map and contributes no outcome to any estimate.

evaluation split averages 18.19 exact supports per prompt; this enumerated support set is not exposed to training or used to initialize replay.

PointMaze. The LM selects a legal adjacent waypoint from the public maze state; a fixed, hash-bound proportional-derivative adapter executes it in the pinned `PointMaze_UMaze-v3` simulator. Wall masking exposes no route or verifier signal. A successful trajectory is keyed by the unique directed barrier corridor it crosses; failures and ambiguities are unkeyed. Evaluation maps are disjoint from interface and warm-start development, and audit routes never enter learning. Appendix E gives the frozen static-domain prompts and PointMaze’s public per-waypoint template.

4 X-MODE GRPO: VERIFIED REPLAY FOR DR.GRPO

x-Mode GRPO is Dr.GRPO plus one persistent, validator-grounded rehearsal channel: verified replay. The learner stores the policy’s own successful outputs after execution, then revisits them after their collection batch. This replay likelihood is the only auxiliary derivative. An output may enter replay only after the same executable validator used for task reward has accepted it and produced its canonical mode key.

Algorithm 1 in Appendix G specifies the exact update.

Verified exemplar bank. For each prompt x , the learner maintains a capacity- M bank $\mathcal{B}_x$, initially empty. A response enters only when its rollout reward is positive and executable canonical validation returns a key $c \neq \perp$. The first policy-generated token sequence for each observed key is retained, up to $M = 16$ keys. Admission never reads exhaustive support, evaluation behavior, a desired mode count, or a reference solution.

Uniform verified likelihood. After the first nonempty bank exists, a recurrent replay schedule selects exactly one prompt-local bank per optimizer update. The implementation uses a fixed round-robin

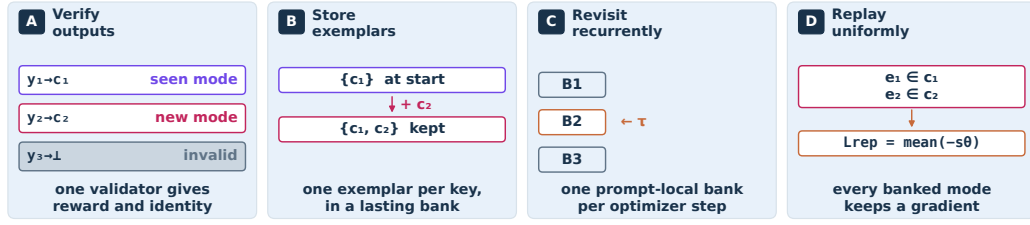


Figure 3: **x-Mode GRPO**. One executable validator supplies both task reward and canonical mode identity (A); the first accepted output for each observed key enters a persistent prompt-local bank (B); a deterministic recurrent schedule revisits one bank per optimizer update (C); and uniform teacher-forced likelihood keeps a gradient on every banked mode, including modes absent from the current on-policy group (D). The compute-matched control runs the same maintenance, scheduling, and teacher-forced traversal with zero applied replay gradient.

cursor so every nonempty bank is revisited. For every retained exemplar e_c in the scheduled bank, the current model computes the length-normalized teacher-forced score

$$s_\theta(e_c | x) = |e_c|^{-1} \log \pi_\theta(e_c | x).$$

The replay loss is

$$L_{\text{replay}}(x) = -\frac{1}{|\mathcal{B}_x|} \sum_{c \in \mathcal{B}_x} s_\theta(e_c | x). \quad (1)$$

Singleton banks are eligible: replay begins as soon as one verified exemplar exists, and each newly observed mode adds one equally weighted exemplar. The categorical reduction in Theorem A.4 shows that the corresponding exact category loss simultaneously creates a barrier against banked-mode extinction and, after full coverage, targets a uniform correct-mode distribution.

The actor minimizes

$$L_{\text{PPO}}(A^{\text{task}}) + \lambda_{\text{replay}} w_{\text{rep}} L_{\text{replay}},$$

where $\lambda_{\text{replay}} = .10$ is a fixed implementation dose and $w_{\text{rep}} = (G - 1)/G^2$ is the fixed row normalization for one replay pseudo-group relative to G on-policy rows. Neither coefficient adapts during training. The theory requires only a positive recurrent effective dose; it does not privilege .10. The matched Dr.GRPO control maintains the same passive bank, scheduler, replay batch, score traversal, and backward-compute envelope but sets the applied replay derivative to exact zero.

4.1 SEMANTIC MAXENT: AN ENTROPY GRADIENT ON VERIFIED OUTCOMES

Verified replay retains what the policy has already found. It supplies no pressure toward outcomes never produced, because a mode absent from the bank contributes no term to Equation 1. Semantic MaxEnt supplies exactly that pressure, and it is derivable rather than heuristic.

Fix a prompt x and let p_θ be the policy-induced distribution over canonical verified outcomes. The map from parameters to outcomes runs sample $\rightarrow$ execute $\rightarrow$ validate $\rightarrow$ canonicalize and is not differentiable, so its Shannon entropy $H(p_\theta)$ admits no pathwise gradient. The log-derivative identity supplies one anyway:

$$\nabla_\theta H(p_\theta) = \mathbb{E}_{a \sim p_\theta} [(-\log p_\theta(a)) \nabla_\theta \log p_\theta(a)], \quad (2)$$

the -1 ordinarily accompanying $-\log p$ vanishing because $\sum_a \nabla_\theta p_\theta(a) = 0$. Entropy ascent is surprisal-weighted policy gradient: no separate entropy objective is needed, only a per-rollout weight. Writing the surrogate with a *detached* plug-in predictor $\hat{p}$ gives $\nabla_\theta S = -\sum_a (\nabla_\theta p_\theta(a)) \log \hat{p}(a)$, which coincides with $\nabla_\theta H$ exactly when $\hat{p} = p_\theta$; detaching is what makes the identity hold, not a convenience.

We estimate $\hat{p}$ from the prompt’s own verified history with an open-set Laplace predictor carrying one reserved UNSEEN bucket, so an outcome never produced still carries mass and the gradient still points toward it (Appendix C). A row is *eligible* only if it is active, parseable, and validator-positive; with surprisal $s_i = -\log \hat{p}(a_i)$ clipped at $C = 5$,

$$A_i^{\text{sem}} = \eta \frac{\min(s_i, C) - \mathbb{E}_{\hat{p}}[\min(s, C)]}{C}, \quad \eta = .10, \quad (3)$$

added to the task advantage after that advantage’s own centering. Every ineligible row receives exact zero and never enters $\hat{p}$ ’s support, so an all-incorrect group is an exact no-op and no wrong answer is rewarded for being unusual. Centering under $\hat{p}$ makes $\mathbb{E}_{\hat{p}}[A^{\text{sem}}] = 0$, and $|A^{\text{sem}}| \leq \eta$, so with the unit reward gap a verified response keeps at least a .90 advantage margin over an incorrect one and the correctness ordering cannot invert.

The two components are complementary by construction: Equation 3 reaches outcomes never produced but has no memory, while Equation 1 has memory but no reach. We measure them separately in Section 7.

5 RELATED WORK

Inference-time compute and verifier-based reasoning RL. Self-consistency, repeated sampling, and verifier-guided search draw a candidate set and select from it [28, 2, 26, 18]; GRPO and Dr.GRPO train the proposal policy itself from group-relative outcome rewards [24, 8, 20]. Dr.GRPO fixes length normalization; our question is orthogonal and upstream: whether RL leaves enough probability *within* equally rewarded correct outcomes for additional inference to remain useful.

Collapse and diversity-aware optimization. RLHF and fine-tuning can trade generalization or $\text{pass}@K$ for narrower outputs [12, 5], and RLVR can lose token entropy and narrow the large- k sampling boundary relative to the base model [4, 31]. Proposed repairs include outcome bonuses, batch penalties, $\text{pass}@K$ objectives, mass-covering divergences, set-level and distribution-matching objectives, and token-level clipping asymmetries [27, 29, 3, 16, 15, 17, 25, 30]. Closest is Song et al. [27], whose 7B competition-math study is stronger external evidence for the general problem; their UCB counts final answers—including incorrect ones—and their bonuses are disabled on all-correct batches, whereas verified replay admits only validator-positive *executed* keys and replays a verified success beyond its collection batch.

Experience replay, archives, and mode-covering objectives. Standard replay buffers retain past transitions so later updates can reuse experience [19, 21]; replay also supports retention in continual learning [23]. Our unit of stored experience is instead a completed successful sequence, indexed by its verified execution mode. Replaying it applies supervised sequence likelihood rather than an off-policy Dr.GRPO task update.

GFlowNets sample in proportion to reward and have fine-tuned LMs toward diverse reasoning chains [1, 11, 32]; quality-diversity search maintains an archive indexed by a hand-designed behavior descriptor [13, 22]; maximum-entropy RL adds entropy regularization over tokens or a known action set [33, 14, 9, 10]; and semantic entropy clusters generations by estimated entailment [7]. Verified replay instead indexes a persistent bank with the validator’s canonical execution key and applies one uniform exemplar-likelihood objective to each scheduled bank. Because the validator computes the key from execution rather than reading a model-asserted label, the equivalence relation is validator-defined; it does not claim that every reasoning strategy is observable.

6 EXPERIMENTAL DESIGN AND EVIDENCE POLICY

6.1 MATCHED TRAINING AND EVALUATION

The clean x-Mode GRPO comparison uses Qwen2.5-0.5B-Instruct at a pinned revision, seeds 43–47, group size 16, $\beta_{\text{KL}} = 0$, rollout temperature 1, and exactly eight passes through each fixed pool. Each static domain trains on 384 prompts and evaluates on a fixed 128-prompt split, giving 3,072 optimizer updates per run. Evaluation and resumable checkpoints occur every 192 updates, the fixed half-pass grid 0, .5, . . . , 8. The original 50 cells are five domains $\times$ two arms $\times$ five paired seeds.

Aligned scale extensions. The same two-arm, eight-pass, 384/128, group-16 design is running on Falcon3-1B-Instruct with seeds 55–59 and on Qwen2.5-3B-Instruct with seeds 70–74. The replay mechanism, capacity, fixed .10 weight, half-pass evaluation grid, and exact-zero derivative control are unchanged. Optimizer recipes are model-specific but fixed before any replay-arm outcome: Falcon uses a constant 2×10^{-7} learning rate, whereas Qwen2.5-3B uses a 10^{-7} cosine schedule with 10% warmup and a 10% minimum-rate floor. Every comparison is within model, domain, and seed;

absolute differences between model families are not treatment effects. These scale cohorts are shown to expose progress, not pooled with the 0.5B cohort.

Within each pair, the control and treatment share data, model, optimizer, rollouts, passive verified bank, deterministic recurrent scheduler, teacher-forced replay traversal, checkpoint cadence, GPU model, and recovery rules. The sole scientific difference is the applied replay derivative: exact zero for matched Dr.GRPO and live uniform verified likelihood for x-Mode GRPO. No other auxiliary derivative is active. The replay weight .10 is a frozen implementation dose, not a selected checkpoint or theoretical constant.

Compute matching. Both arms perform identical rollout generation, verified-bank maintenance, recurrent scheduling, teacher-forced exemplar scoring, and backward traversal. The control sets the resulting replay derivative to exact zero; the treatment applies it. The comparison therefore changes the learning signal without changing its compute envelope. Relative to ordinary Dr.GRPO without this matched traversal, verified replay adds teacher-forced scoring of at most 16 previously generated exemplars per update. Terminal reporting will include measured throughput rather than infer the overhead from the loss coefficient.

PointMaze stratum. Qwen2.5-0.5B uses the same paired eight-pass protocol on 384 training and 128 evaluation maps, with seeds 43–47 and a shared frozen 72-update warm start learned on disjoint maps. A replay exemplar is one successful route: treatment applies its mean selected-action log likelihood, while control performs the same traversal with an exact-zero derivative. All other auxiliary signals are off, so this remains the same Dr.GRPO-versus-x-Mode GRPO test but a separate estimator from the static domains. The parallel Falcon3-1B stratum uses seeds 55–59 and a shared Falcon warm start; four control cells, but no replay cell, had registered evaluation in the dated snapshot. No Qwen2.5-3B PointMaze stratum is scheduled.

6.2 EVIDENCE POLICY

Each environment is analyzed separately. The primary estimands are the paired seed differences, x-Mode GRPO minus matched Dr.GRPO, at pass 8 for `distinct@8` and `pass@8`. The secondary trajectory estimand is trapezoidal AUC over the complete half-pass grid. The final report will include all five paired seed differences and their mean and range. We do not pool environments, model families, or interface strata, and never select a best checkpoint. For the five static domains, descriptive `pass@1` uses greedy decoding; sampled `mean@8` and excess multiplicity are also reported descriptively.

PointMaze uses the same two primary estimands and trajectory summary but has no greedy full-program endpoint: its actions are sequential and evaluated under a common-random-number interactive protocol. The five-domain 0.5B cohort becomes final only after all 50 pass-8 endpoints pass the frozen audit; each scale cohort and the PointMaze cohort becomes final only after every registered cell in it passes its prospective audit. Until then, the dated values are explicitly interim. The evidence ledger, state-integrity rules, exclusions, dataset auditing, and cohort provenance are bound in Appendices L and F.

7 RESULTS

Snapshot boundary. Figure 4 freezes the registered evaluations available at 22:04 EDT on August 9, 2026: 60/60 Qwen2.5-0.5B including PointMaze, 58/60 Falcon3-1B, and 16/50 Qwen2.5-3B cells with at least one registered evaluation. Runs continue after this cutoff, and Table 1 reports the deepest checkpoint shared by both arms within it.

Interim directions so far. At the deepest paired checkpoint, x-Mode GRPO has a non-negative `pass@8` point estimate in all 17 currently evaluable model–domain panels, strictly positive in 16; its `distinct@8` estimate is positive in 16 of them. The exception in both cases is Falcon3-1B PointMaze, whose deepest paired checkpoint is early in training on five seeds, where `pass@8` is exactly level and `distinct@8` is lower by .016. Among the 15 static panels with a greedy endpoint, its `pass@1` is higher in 13 and lower on Falcon3-1B PantryPlan (−.060) and Qwen2.5-3B PantryPlan (−.016). Sampled `mean@8` is higher in 12/17 panels and lower on Falcon3-1B PantryPlan (−.087), Falcon3-1B PointMaze (−.003), and the Qwen2.5-3B Graph (−.017), Countdown (−.081),

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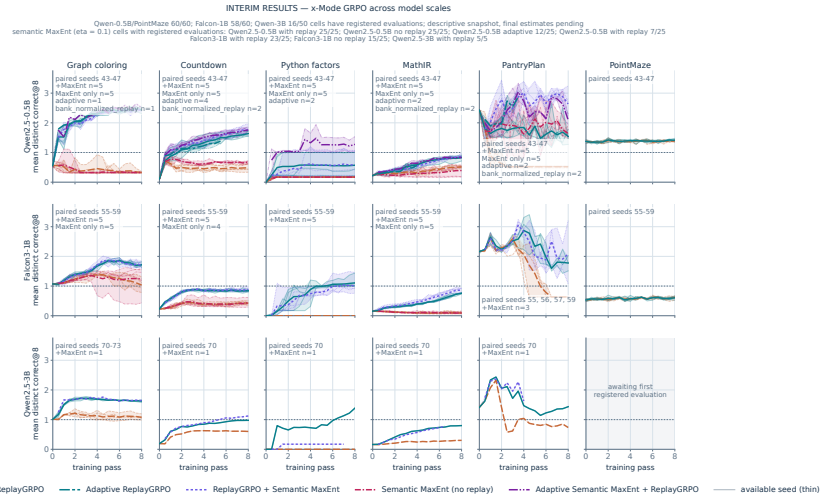


Figure 4: **Interim x-Mode GRPO trajectories across model scales.** The metric is mean distinct@8; columns are environments and rows are model families. Curves and shaded min–max ranges use only seeds with both matched Dr.GRPO and x-Mode GRPO evaluations at each checkpoint. The three semantic-MaxEnt arms are each paired against the arm they are added to: + *semantic MaxEnt* and *adaptive semantic MaxEnt* against x-Mode GRPO, *semantic MaxEnt*, *no replay* against matched Dr.GRPO, so a panel note reports that arm’s own paired seed count. Gray regions are unfinished; blank panels have no registered evaluation or no scheduled cohort. As everywhere in the paper, no domain is pooled with any other. This dated snapshot is descriptive, uses no imputation or checkpoint selection, and will be replaced after the registered cohorts complete.

Model	Domain	n @pass	Dr.GRPO				x-Mode GRPO			
			pass@1	pass@8	mean@8	distinct@8	pass@1	pass@8	mean@8	distinct@8
Qwen2.5-0.5B	Graph coloring	5@8.0	.323	.323	.323	.325	.520	.969	.404	2.438
	Countdown	5@8.0	.464	.480	.465	.487	.630	.672	.597	1.637
	Python factors	5@8.0	.172	.172	.172	.172	.519	.528	.507	.571
	MathIR	5@8.0	.466	.512	.464	.513	.708	.796	.688	.829
	PantryPlan	5@8.0	.522	.522	.522	.522	.569	.729	.545	1.539
	PointMaze	5@8.0	—	.781	.404	1.389	—	.789	.409	1.414
Falcon3-1B	Graph coloring	5@8.0	.305	.656	.300	1.034	.366	.809	.334	1.717
	Countdown	5@8.0	.277	.328	.268	.420	.405	.636	.340	.841
	Python factors	5@8.0	.000	.000	.000	.000	.872	.979	.722	1.107
	MathIR	5@8.0	.073	.095	.068	.097	.495	.736	.292	.756
	PantryPlan	3@8.0	.633	.633	.633	.633	.573	.767	.546	1.776
	PointMaze	5@8.0	—	.511	.117	.620	—	.511	.114	.605
Qwen2.5-3B	Graph coloring	4@8.0	.357	.684	.364	1.077	.424	.833	.347	1.626
	Countdown	1@8.0	.578	.602	.571	.602	.602	.746	.490	.977
	Python factors	1@8.0	.000	.000	.000	.000	.688	.762	.644	1.387
	MathIR	1@8.0	.195	.301	.206	.301	.633	.762	.626	.799
	PantryPlan	1@8.0	.609	.648	.599	.734	.594	.742	.561	1.436

Table 1: **Four metrics at the deepest available paired checkpoints in the frozen interim snapshot.** Columns compare matched Dr.GRPO with x-Mode GRPO, whose only auxiliary derivative is verified replay. The historical multi-component treatment is not included. Entries are paired-seed means; n @pass gives the paired-seed count and deepest common training pass. Columns report greedy pass@1, sampled pass@8, sampled mean@8, and sampled distinct@8. PointMaze’s actions are sequential, so it has no greedy full-program endpoint, hence “—.” Unavailable panels are omitted rather than encoded as zero. Unequal n and nonterminal passes make this a progress table, not a final estimate or uncertainty analysis. Pale green marks an x-Mode GRPO value that is strictly higher than its matched Dr.GRPO value; ties, decreases, and unavailable cells are unshaded.

and PantryPlan (−.038) checkpoints. For distinct@8, provisional differences range from −.016 modes on that early Falcon PointMaze panel to +2.112 on Qwen2.5-0.5B Graph coloring.

Scope. These are descriptive checkpoint summaries rather than registered terminal estimates; panels differ in seed count and no domain is pooled with another. No job was stopped and no checkpoint

was selected because of the snapshot, and superseded multi-component results remain outside the evidence for x-Mode GRPO.

7.1 ADDING SEMANTIC MAXENT TO VERIFIED REPLAY

Section 4.1 separates discovery from retention; two further registered arms per family test that separation, each differing from its comparator by one applied derivative. *Replay* + *MaxEnt* adds Equation 3 to x-Mode GRPO; *MaxEnt only* adds it to matched Dr.GRPO with the replay derivative at exact zero. Each inherits its comparator’s data, seeds, schedule, decoding, placement, and code, with $\eta = .10$ fixed in advance and never tuned against an outcome.

Table 2 in Appendix C reports the full grid of paired seed differences in `distinct@8` at pass 8. A third arm, also drawn in Figure 4, replaces the fixed η with a controller holding realized semantic pressure at a fixed fraction of task-advantage RMS; its mechanism gate is met and its outcome is not yet readable (Appendix B.3).

What the two arms show. Added on top of verified replay, semantic MaxEnt increases `distinct@8` in six of the nine currently evaluable model–domain cells, in both families. In most it is a refinement beside replay’s own effect — replay lifts Qwen2.5-0.5B Countdown from .487 to 1.637 modes and MaxEnt adds a further .115 — but on Qwen2.5-0.5B PantryPlan it adds .922 modes, from 1.558 to 2.480, which is comparable to what replay contributes there.

The MaxEnt-only arm separates the mechanisms: it moves `distinct@8` by +.184 on Countdown but by $\pm .000$ on Python factors and +.004 on Graph coloring, where x-Mode GRPO itself gains .399 and 2.113. Discovery pressure alone does not reproduce retention. On MathIR the term is $-.121$ alone but +.035 on top of replay, consistent with discovery paying off once something holds what it finds. PantryPlan is the exception in the other direction: the term is large both alone (+1.186 on two seeds) and on top of replay (+.922), which is what one expects where the verified outcome space is wide enough that discovery, not retention, is the binding constraint.

8 CONCLUSION

Two distinctions organize the design. *Identity*: accepted execution defines a mode. *Retention versus discovery*: replay protects an observed, banked mode but does not guarantee that an unobserved mode will be found.

Limitations. The registered program covers five synthetic static-response domains, one synthetic-map MuJoCo PointMaze environment, and three model scales or families, with five paired seeds planned per stratum. Exact mode identity is unusually available. PointMaze delegates low-level force control to a fixed PD adapter, so it tests language-level waypoint choice rather than end-to-end motor learning. Capacity-16 banks can omit later discoveries, the fixed .10 replay dose is neither ablated nor claimed optimal, and the same holds for the .10 semantic coefficient. The theoretical guarantee assumes a recurrently replayed fixed bank in a categorical model: a conditional retention result, not a discovery guarantee or a convergence proof for a language model.

What it adds up to. MODEBENCH provides execution-grounded measurement of correct-mode collapse across static and interactive domains, and the categorical analysis gives x-Mode GRPO a conditional post-discovery retention guarantee through verified replay. Every currently paired interim breadth panel points toward x-Mode GRPO, and adding an entropy gradient on verified outcomes adds to it in most panels. These interim point estimates support no neural performance claim until each registered pass-8 stratum is complete and audited.

ETHICS STATEMENT

The benchmark and experiment-specific data are synthetic; this study collects no human-subject or personal data. More verified modes are not inherently better: a distinct execution may be risky or costly even when it passes the validator, and `distinct@K` does not measure downstream value. Systems that evaluate or optimize breadth should report correctness, invalidity, and application-specific costs alongside mode counts.

REPRODUCIBILITY STATEMENT

Every quantitative claim is bound to a frozen artifact. The clean comparison pairs each domain and seed within GPU model; physical device identity need not match. Because GPU model can change sampled tokens, the audit rejects a pair whose GPU models differ. Section 6 and Appendix L give the matched protocol, the complete hyperparameter set, and the evidence ledger that fixes what each comparison may claim; Appendix F records SHA-256 identities for all training and evaluation splits. Comparison protocols were registered before results were observed and are included as `paper/preregistration/*.md`. Bank, scheduler, optimizer, data-cursor, and request-stream state are checkpointed atomically, and integrity violations—non-finite values, state mismatch, missing terminal markers, or proposal leakage into PPO—fail closed rather than degrade silently. Terminal figures are regenerated from hash-bound artifacts by named scripts listed in Appendix L, and rendering aborts if a displayed endpoint does not reproduce its frozen aggregate. The interim Figure 4 is deliberately frozen together with its full JSON provenance rather than silently advancing on each build. Two superseded cohorts are excluded by machine-readable audits written before their outcomes were known.

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A WHY GRPO COLLAPSES AND X-MODE GRPO RETAINS MODES

Main-body link. This appendix supplies the formal claims invoked in Sections 4 and 7.

This appendix separates two questions that are easy to conflate: whether a binary-reward group update preserves ratios among already correct modes, and whether x-Mode GRPO preserves modes that the policy has already produced and the validator has accepted. The first admits an unconditional answer in a finite categorical model. The second necessarily requires a coverage assumption: no method whose bank contains only executed, validator-positive objects can protect a mode that is never sampled or constructed. We therefore prove support preservation first and state full no-collapse as a coverage corollary. Accordingly, the positive theorem explains the stability mechanism after discovery; it does not prove that x-Mode GRPO will establish its own coverage premise in a language model.

A.1 CATEGORICAL REDUCTION AND THE GRPO MEAN UPDATE

Fix one prompt. Let the finite response categories be $\mathcal{A} = \mathcal{C} \dot{\cup} \mathcal{N}$, where the $m \geq 2$ categories in $\mathcal{C}$ are correct execution modes and the $n \geq 1$ categories in $\mathcal{N}$ are incorrect. Let $p_a = \text{softmax}(z)_a$, $P = \sum_{c \in \mathcal{C}} p_c$, and $q_c = p_c/P$. Thus q is exactly the conditional correct-mode distribution $p_\theta^+(\cdot | x)$ of Section 2, and P is the correct mass $P_\theta^+(x)$.

Draw a group $A_1, \dots, A_G \stackrel{\text{iid}}{\sim} p$ with $G \geq 2$, write $R_i = \mathbf{1}[A_i \in \mathcal{C}]$, and let $M = \sum_i R_i$. Both GRPO variants can be written, at the on-policy point where the PPO ratio equals one, as

$$\hat{g}_G = \frac{1}{G} \sum_{i=1}^G w(M) \left(R_i - \frac{M}{G} \right) \nabla_z \log p_{A_i}. \quad (4)$$

For Dr.GRPO, $w(M) = 1$ after absorbing the shared positive length scale. For binary-reward GRPO, $w(M)$ is the reciprocal group reward standard deviation on mixed groups. In either case $w(\ell) > 0$ for $\ell \in \{1, \dots, G-1\}$; its value on all-equal groups is irrelevant because the centered advantage is zero. We take explicit reference KL to be zero, as in the experiments.

Lemma A.1 (A group baseline does not remove frequency amplification). *For $0 < P < 1$, the expected group update is*

$$\mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P, \quad c_G(P) = \frac{\mathbb{E}[w(M)M(G-M)]}{G^2 P(1-P)} > 0. \quad (5)$$

For Dr.GRPO, $c_G(P) = (G-1)/G$. Hence reward centering and binary reward standardization change the speed, but not the expected direction, of the binary-reward policy gradient.

Proof. Condition on the complete binary reward vector, whose sum is M . The score identities give

$$\mathbb{E}[\nabla_z \log p_{A_i} | R_i = 1] = \nabla_z \log P, \quad \mathbb{E}[\nabla_z \log p_{A_i} | R_i = 0] = \nabla_z \log(1-P).$$

There are M correct rows with centered reward $(G-M)/G$ and $G-M$ incorrect rows with centered reward $-M/G$. Therefore the conditional expectation of the unaveraged sum in (4) is

$$w(M) \frac{M(G-M)}{G} (\nabla_z \log P - \nabla_z \log(1-P)) = w(M) \frac{M(G-M)}{GP(1-P)} \nabla_z P.$$

The outer factor $1/G$ and expectation over $M \sim \text{Binomial}(G, P)$ yield (5). For $w \equiv 1$, $\mathbb{E}[M(G-M)] = G(G-1)P(1-P)$, proving the Dr.GRPO value. $\square$

A.2 WINNER-TAKE-ALL COLLAPSE UNDER GRPO AND DR.GRPO

Consider the infinitesimal expected-update flow $\dot{z} = \mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P$. This is the clean mean-field limit of fresh on-policy groups and the first-order PPO update. It deliberately removes sampling noise; the result below therefore identifies a structural instability rather than blaming finite batches.

Theorem A.2 (Generic binary-reward GRPO mode collapse). *Start from finite logits with $0 < P(0) < 1$. If the conditional correct-mode distribution $q(0)$ has a unique largest coordinate c^* , then*

$$P(t) \longrightarrow 1, \quad q_{c^*}(t) \longrightarrow 1, \quad q_c(t) \longrightarrow 0 \quad (c \neq c^*).$$

Thus both GRPO and Dr.GRPO converge to a single correct execution mode for generic initial conditions, even though every mode in $\mathcal{C}$ has exactly the same reward.

Proof. For a correct mode c and incorrect category a ,

$$\dot{z}_c = c_G(P)p_c(1 - P), \quad \dot{z}_a = -c_G(P)p_aP.$$

Also $\dot{P} = c_G(P)\|\nabla_z P\|_2^2 > 0$. If P converged to an interior value, continuity and positivity of c_G , together with

$$\|\nabla_z P\|_2^2 \geq P^2(1 - P)^2 \left(\frac{1}{m} + \frac{1}{n} \right),$$

would keep $\dot{P}$ bounded away from zero, a contradiction. Hence $P \rightarrow 1$.

Define the effective optimization time

$$\tau(t) = \int_0^t c_G(P(s))P(s)(1 - P(s)) ds.$$

If $\tau(\infty) < \infty$, every correct and incorrect logit would have finite total variation, because the absolute velocity of each is at most $\dot{\tau}$. All logits would then converge to finite values, contradicting $P \rightarrow 1$. Therefore $\tau(t) \rightarrow \infty$.

On the correct simplex, changing from t to τ gives the replicator flow

$$\frac{dq_c}{d\tau} = q_c \left(q_c - \sum_{d \in \mathcal{C}} q_d^2 \right), \quad \frac{d}{d\tau} \log \frac{q_c}{q_d} = q_c - q_d. \quad (6)$$

The initially unique maximum remains the unique maximum, and every one of its log-ratios increases. The Lyapunov quantity $\sum_c q_c^2$ is nondecreasing, since its derivative is twice the variance of the coordinate values under q . Every limit point of (6) is therefore uniform on its positive support. Because $q_{c^*} \geq 1/m$ and its ratio to every other coordinate stays strictly greater than one, no multi-coordinate uniform limit is possible. The only limit point is the vertex $q_{c^*} = 1$. $\square$

Corollary A.3 (Ties are nongeneric). *If exactly $k < m$ correct modes tie for the largest initial probability, symmetry preserves the tie and the flow converges to the uniform distribution on those k modes while eliminating the rest. Only the exactly uniform initialization over all m correct modes avoids collapse. Any arbitrarily small perturbation that creates a unique maximum reduces to Theorem A.2.*

Finite groups make this instability operationally harsher: a mode absent from a group has no score carrier, all-correct groups have zero task advantage, and sampling noise almost surely breaks exact ties. These facts are not needed for the theorem.

A.3 WHY UNIFORM VERIFIED REPLAY PREVENTS EXTINCTION

Let $\mathcal{B} \subseteq \mathcal{C}$ be a nonempty fixed bank of k modes that the policy has produced and the validator has accepted. Uniform verified replay uses one exemplar per banked mode and, in the categorical reduction, minimizes

$$R_{\mathcal{B}}(z) = -\frac{1}{k} \sum_{b \in \mathcal{B}} \log p_b. \quad (7)$$

This is ordinary cross-entropy against the distribution that assigns mass $1/k$ to each banked mode and zero elsewhere. It is not conditioned on the bank: decreasing (7) can raise total bank mass as well as redistribute mass within the bank.

Absorb a fixed round-robin replay frequency and loss scale into an effective dose $\rho > 0$. The prompt-isolated categorical mean flow for binary-reward GRPO plus verified replay is

$$\dot{z} = c_G(P)\nabla_z P - \rho \nabla_z R_{\mathcal{B}}(z). \quad (8)$$

Let $\Psi(P) = \int_0^P c_G(s) ds$. For the finite-group estimators in Lemma A.1, c_G has a finite positive extension to $[0, 1]$; in particular, Dr.GRPO has $c_G = (G - 1)/G$. Define

$$F_B(z) = \rho R_B(z) - \Psi(P(z)). \quad (9)$$

Then (8) is exactly gradient flow $\dot{z} = -\nabla_z F_B$.

Theorem A.4 (Verified replay prevents banked-mode extinction). *Suppose that after time T the nonempty verified bank $\mathcal{B}$ is fixed, fits within replay capacity, and receives a fixed positive effective replay dose ρ . Under (8), define*

$$C_T = R_B(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho}.$$

Then for every $b \in \mathcal{B}$ and every $t \geq T$,

$$p_b(t) \geq \exp(-kC_T) > 0. \quad (10)$$

Thus no verified mode in a recurrently replayed bank can vanish, even when the bank does not cover every correct mode.

Proof. Along the flow,

$$\frac{d}{dt} F_B(z(t)) = -\|\nabla_z F_B(z(t))\|_2^2 \leq 0.$$

Since $P(t) \leq 1$ and Ψ is increasing,

$$R_B(z(t)) \leq R_B(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho} = C_T.$$

Every $-\log p_b$ is nonnegative and their sum is kR_B . Therefore $-\log p_b(t) \leq kC_T$ for every banked mode, which is exactly Equation (10). $\square$

The bound can be conservative when ρ is small, but its qualitative content is sharp: any positive recurrent replay dose creates a log barrier against extinction. The numerical dose controls finite-time strength, not which distribution the categorical objective targets.

Corollary A.5 (x-Mode GRPO no-collapse guarantee after verified coverage). *Assume the conditions of Theorem A.4 and that by time T the bank covers the full correct support, $\mathcal{B} = \mathcal{C}$. Let $u_c = 1/m$ on $\mathcal{C}$. Then the categorical mean flow converges to*

$$P(t) \longrightarrow 1, \quad q(t) \longrightarrow u.$$

Consequently

$$\min_{c \in \mathcal{C}} q_c(t) \longrightarrow \frac{1}{m}, \quad \mathcal{H}(q(t)) \longrightarrow \log m,$$

and conditional correct-mode collapse is impossible.

Proof. When $\mathcal{B} = \mathcal{C}$,

$$R_C = \log m - \log P + \text{KL}(u||q). \quad (11)$$

Thus one uniform verified-likelihood loss already contains both a verified-mass term $-\log P$ and a within-support uniformity term $\text{KL}(u||q)$. The potential (9) becomes

$$F_C = \rho \log m + \rho \text{KL}(u||q) - \rho \log P - \Psi(P).$$

For fixed P , its unique minimizer is $q = u$. Its derivative in P is $-\rho/P - c_G(P) < 0$, so its unique limiting distribution is $P = 1, q = u$.

For completeness, Theorem A.4 keeps every correct coordinate away from zero. The decreasing potential gives $\int_T^\infty \|\nabla F_C\|_2^2 dt < \infty$; bounded softmax derivatives imply $\nabla F_C \rightarrow 0$. At any limiting distribution the correct-coordinate stationarity equations are

$$0 = \rho(p_c - 1/m) - c_G(P)p_c(1 - P),$$

so all p_c are equal. Substituting $p_c = P/m$ gives $0 = -(1 - P)(\rho + c_G(P)P)/m$, hence $P = 1$. Each incorrect coordinate satisfies $0 = p_a(\rho + c_G(P)P)$ and therefore has $p_a = 0$. The limiting distribution is unique, proving convergence. $\square$

Remark A.6 (Exact scope of the x-Mode GRPO guarantee). The theorem is a conditional post-discovery retention result, not an end-to-end discovery theorem. Without full coverage it protects every mode in $\mathcal{B}$ but says nothing about unobserved modes in $\mathcal{C} \setminus \mathcal{B}$; capacity 16 can also truncate the protected support. The categorical proof treats each stored exemplar as its execution category and uses $\log p_i$, whereas the language-model implementation uses a length-normalized teacher-forced exemplar score. Cross-prompt parameter interference, stochastic finite steps, and nonconvex representation are outside the mean-flow result and must be tested empirically. These limitations motivate the matched five-domain x-Mode GRPO comparison rather than a stronger neural network claim.

B SUPPORTING MEASUREMENTS

Four measurements taken from the registered cohorts’ own telemetry. None requires a new experiment; each is re-derived from the run logs of cells already reported.

B.1 WHY MATCHED DR.GRPO HAS ALMOST NOTHING TO LEARN FROM

Dr.GRPO centres advantages within a rollout group, so a group whose members all receive the same reward contributes *exactly zero* gradient. Only a *mixed* group — at least one correct and at least one incorrect member — produces any learning signal at all. The fraction of mixed groups is therefore the fraction of updates that do any work.

Measured over all 3,072 updates of the Qwen2.5-0.5B cohort, five seeds per cell:

Domain	matched Dr.GRPO	x-Mode GRPO
Graph coloring	.144	.954
Countdown	.140	.456
Python factors	.015	.185
MathIR	.364	.453
PantryPlan	.123	.789

On Python factors the control spends 98.5% of its updates on groups that cannot move the policy. The effect sharpens with scale: the Qwen2.5-3B control on the same domain logged **7 mixed groups in 3,073 updates**, while its paired x-Mode GRPO run logged 843. That is the mechanism behind the zero entry in Table 1: the baseline does not fail to optimize, it has no gradient to optimize with, and verified replay supplies one by keeping rare successes in the batch.

B.2 COLLAPSE IS REGRESSION, NOT MERELY A SMALLER GAIN

Every panel shares a pass-0 reference, since both arms start from the same model. In 6 of the 17 evaluable panels the matched Dr.GRPO arm finishes *below* that reference: Qwen2.5-0.5B Graph coloring .561 $\rightarrow$.325 and PantryPlan 2.426 $\rightarrow$.522; Falcon3-1B Graph coloring 1.070 $\rightarrow$ 1.034, MathIR .154 $\rightarrow$.097, and PantryPlan 2.162 $\rightarrow$.633; Qwen2.5-3B PantryPlan 1.412 $\rightarrow$.734. Training on verified reward alone destroys correct-mode breadth the base model already had. x-Mode GRPO ends below its pass-0 reference in 2 of 17 panels, both PantryPlan.

B.3 A FIXED COEFFICIENT DOES NOT DELIVER A FIXED DOSE

Section 4.1 fixes $\eta = .10$. The *realized* pressure, measured as the ratio of applied semantic-advantage RMS to task advantage RMS over all 3,072 updates of seed 43 in the semantic-MaxEnt arm, is not fixed at all: Countdown 3.71%, Graph coloring 1.64%, Python factors 1.88%, MathIR 1.02%. Within Python factors the per-seed eligible fraction spans .21–.79 and the semantic RMS spans .00031–.00250, an eightfold range at an identical nominal coefficient. The cause is structural: the centered surprisal of Equation 3 depends on how many verified modes the prompt’s bank happens to hold. A coefficient is therefore not a dose.

A registered follow-up adapts η per run to hold this ratio at a fixed target. Its first cohort establishes a constraint worth recording: with η bounded above by .40 — the largest value that still leaves a

verified response a .60 advantage margin under a unit reward gap, and so the largest that preserves the correctness ordering of Appendix C — a 5% target is unreachable in MathIR and Python factors, whose controllers sat at that ceiling for 97–98% of updates while realizing only 2.8% and 1.7%. Equalizing semantic pressure across domains is bounded by the safety constraint, not by the controller.

A second cohort re-derives the target from what those saturated cells sustained rather than from the bound: the lowest ratio any ceiling-pinned cell held was .0170, so $\rho = .015$ is registered with η 's bound and every other setting unchanged. Its mechanism gate — read before any outcome, as registered — is met on the four domains so far in progress. Realized ratio is .0131–.0164 against the .0150 target, inside 13% where the registered criterion allowed a factor of two; η sits at a bound on 0–5.8% of applied updates against a 20% allowance; no domain is frozen above 57% against an 80% threshold; and mean response length is within 5.4% of the fixed arm with no-EOS at zero in every cell. The coefficients that hold one common ratio span .033–.144, which is this section's own point from the other side: equal realized pressure requires unequal coefficients. Python factors, whose .0170 set the target, has not yet started, so reachability is established on four domains of five, and the arm's registered outcome — its paired difference against the fixed arm at pass 8 — is not yet readable.

B.4 POINTMAZE IS WARM-STARTED AND NEAR-STATIONARY

The five static domains train from the base instruct model. PointMaze cannot: neither base model drives a maze at all, so both PointMaze cohorts begin from a supervised warm start — Qwen2.5-0.5B from a checkpoint trained for 72 optimizer steps to .720 train accuracy on a disjoint map set, Falcon3-1B from its own 72-step warm start at .528. The warm-start data identity is recorded in the evaluation split's exclusion list, so the sets are disjoint by construction.

The consequence for reading Table 1: over eight passes the Qwen2.5-0.5B PointMaze arms move by +.016 and +.041 modes from a shared pass-0 value of 1.373, against −.236 and +1.876 for Graph coloring on the same row. Both PointMaze arms end within .025 modes of where the warm start put them. Its cells are not evidence of transfer in either direction, and the directional counts in Section 7 would be better read over the 15 base-initialized static panels alone.

C SEMANTIC MAXENT: ESTIMATOR AND PROPERTIES

Section 4.1 states the entropy-gradient identity and the applied advantage. This appendix gives the predictor it is evaluated under and the properties that follow.

Open-set prompt-local predictor. For prompt x , let the explicit support be the union of previously verified canonical outcomes and the leave-one-out verified peers of the current group, with K distinct keys whose counts sum to N . With Laplace pseudocount $\alpha = 1$ and one reserved UNSEEN bucket,

$$\hat{p}(a) = \frac{n(a) + \alpha}{N + \alpha(K + 1)}, \quad \hat{p}(\text{UNSEEN}) = \frac{\alpha}{N + \alpha(K + 1)}. \quad (12)$$

The reserved bucket is what makes the estimator open-set. A closed-set estimator over K observed keys assigns zero mass to everything unobserved, its surprisal diverges, and the gradient toward genuinely new outcomes is undefined. Equation 12 keeps that mass finite and positive, so discovery pressure is well defined before a second mode exists.

Exact centering. The expectation in Equation 3 is taken under the same detached $\hat{p}$ that scores the row, so $\mathbb{E}_{\hat{p}}[A^{\text{sem}}] = 0$ identically rather than approximately. The term therefore redistributes advantage among verified rows without shifting the objective level, and adds no net drift to the update.

Bounded pressure. The centered clipped surprisal lies in $[-1, 1]$, hence $|A^{\text{sem}}| \leq \eta$. Every task in this paper uses a unit correct/incorrect reward gap, so at $\eta = .10$ a verified response retains at least a .90 advantage margin over an incorrect one. Semantic MaxEnt can reorder verified outcomes among themselves; it cannot make an incorrect output preferred.

The clip is part of the scale. Replacing s by $\min(s, C)$ makes this a truncated-entropy surrogate rather than exact entropy ascent, and C and η act jointly: at $C = 5$ an outcome never seen after 200 prior verified rows saturates the clip and earns nearly the full η , while at $C = 50$ the same event earns roughly a tenth of that. Reported coefficients are therefore meaningful only alongside their clip.

Model	Domain	n	(replay+MaxEnt) – replay	MaxEnt – Dr.GRPO
Qwen2.5-0.5B	PantryPlan	4	+.922	+1.186
	Countdown	5	+.115	+.184
	MathIR	5	+.035	–.121
	Python factors	5	+.016	±.000
	Graph coloring	5	–.022	+.004
Falcon3-1B	MathIR	5	+.108	–
	Countdown	5	+.048	–
	Graph coloring	5	–.006	–
	Python factors	5	–.091	–

Table 2: **Paired distinct@8 differences at pass 8.** Each row is a within-seed difference against that arm’s own comparator, since the two arms add the same term to different bases. PantryPlan’s cells are re-runs: its canonical-action outcome key was initially unparse by the semantic estimator, leaving the term inert, and these replace those runs. Its MaxEnt-only column rests on two seeds. The Falcon PantryPlan and MaxEnt-only cells are still in flight.

Eligibility. Only active, parseable, validator-positive rows are eligible, and ineligible rows never enter $\hat{p}$ ’s support. This is what prevents the term from rewarding diverse failure: novelty among wrong answers is invisible to the estimator. It also means the term is exactly inert on any task whose canonical outcome key the estimator cannot parse, which is a property to verify per domain rather than assume.

D EXTERNAL PYTHON VERIFICATION CONTRACT

Main-body link. This appendix expands the Python Factors contract in Section 3 and its limitation in Section 8.

A Python response is parsed in expression mode and must be a lambda with one argument named n . Accepted nodes are integer constants, n , bounded arithmetic, comparisons, Boolean operators, unary operators, and conditional expressions. Calls, imports, attributes, comprehensions, containers, subscripts, assignment, loops, and every other identifier are rejected before execution. Candidate length and AST size are bounded.

The parent sends restricted source and trusted JSON task data to a persistent external interpreter over JSON lines. The worker reparses, removes builtins, evaluates the lambda, and invokes it on all frozen inputs under a per-request timer. It returns only validity, the integer vector, and its key. Timeout, broken pipe, malformed output, non-integer values, or divisor failure returns no key. The parent can restart a failed worker.

For inputs [6, 10, 15],

```
lambda n: 2 if n % 2 == 0 else 3
```

returns (2, 2, 3), while

```
lambda n: 3 if n % 3 == 0 else 5
```

returns (3, 5, 3). Both verify and receive different behavior keys.

E DOMAIN PROMPTS

Main-body link. This appendix reproduces the exact prompt the policy receives in each of the five MODEBENCH domains of Section 3 and Table 3.

Every domain is single-turn: one system message, one user message, and no few-shot examples. The user message is the dataset `problem` field; the wrapper is the chat template the launcher exported through `OAT_ZERO_PROMPT_TEMPLATE`, defined in `src/oat_drgrp/ templates.py`. Both arms of every domain use the identical string, and the same template is applied at training and at evaluation, so the prompt is never a difference between arms. Each block below is the first row of that cohort’s frozen evaluation split, rendered by that template; `ops/check_paper_domain_prompts.py` regenerates these blocks from the datasets and fails closed if the typeset text drifts from them. Lines longer than the column are wrapped here with a two-space continuation indent; joining each indented continuation to its predecessor with one space recovers the string byte for byte, and the generator asserts that round trip. The only real newlines are the ones inside the `PantryPlan` pantry listing.

PointMaze is not one of these single-turn blocks. Its policy receives a fresh prompt after every executed waypoint. The public template in `src/oat_drgrp/point_maze_waypoint_ policy.py` renders the map, current and previous cells, goal, continuous position and velocity, remaining horizon, and only the adjacent free moves; both arms receive identical state and option strings. Every observed state–action transition is hash-bound.

E.1 GRAPH COLORING

Template `qwen_boxed`; 384 training and 128 evaluation rows in `var/data/graph_ coloring_modebench_v2`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Color this graph with vertices 1 through 6. The undirected edges are: (1,2), (1,3), (3,5),
(3,6), (4,6), (5,6). Assign each vertex one color from {1, 2, 3} so that no edge has the
same color on both endpoints. A partial coloring is 2??1?2, where ? means uncolored.
Keep every shown digit fixed. There are exactly 3 question marks. Return exactly one
final answer and no explanation: exactly 3 digits, each one 1, 2, or 3, for the missing
positions from left to right, inside \boxed{}.<|im_end|>
<|im_start|>assistant
```

E.2 COUNTDOWN

Template `qwen_boxed`; 384 training and 128 evaluation rows in `var/data/exact_ countdown_easy3_probe`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Using the numbers [3, 6, 9], create an arithmetic expression that equals 18. Use each
given number exactly once. You may use +, -, *, /, and parentheses. Return exactly one
final answer and no explanation: the expression inside \boxed{}.<|im_end|>
<|im_start|>assistant
```

E.3 PYTHON FACTORS

Template `qwen_boxed`; 384 training and 128 evaluation rows in `var/data/python_factor_ modebench_v1`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Write one pure Python function with the exact form lambda n: EXPR. An external Python tool
will call it once for each n in [18, 82, 91, 93]. For every call, return an integer d
satisfying 1 < d < n and n % d == 0. Different valid return vectors are different
solution modes. You may use integer literals, n, +, -, *, /, %, comparisons, Boolean
```

```

operators, and conditional expressions; calls, imports, attributes, containers, and
other names are forbidden. Return exactly the one-line lambda inside \boxed{} and no
explanation.<|im_end|>
<|im_start|>assistant

```

E.4 MATHIR ACTION MENU

Template `qwen_boxed`; 384 training and 128 evaluation rows in `var/data/mathir_action_menu_v1`.

```

<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
MathIR linear-menu-v1. Solve for x by choosing an executable action sequence. Every
selected action is applied exactly to BOTH sides of the current equation, followed by
exact simplification. Bindings: a=2, b=-9, c=-1. Initial equation: x/a + b = c. Action
menu: A means div(a); B means add(b); C means sub(b); D means sub(c); E means
sub(mul(a,b)); F means mul(a). Choose 1-4 action IDs. Illegal operations, repeated
states, and paths that do not finish with x isolated are rejected. Return only the IDs
separated by semicolons inside the answer box; for example formatting only: \boxed{A;C}.
Do not return x's numeric value.<|im_end|>
<|im_start|>assistant

```

E.5 PANTRYPLAN

Template `qwen_pantry_support_mask`; 384 training and 128 evaluation rows in `var/data/pantry_plan_modebench_v2`.

```

<|im_start|>system
Return only six binary digits and no other text. Each digit maps to the corresponding
Pantry row in printed order. A trusted environment will choose exact quantities on
precisely the selected rows.<|im_end|>
<|im_start|>user
Create one feasible high fiber snack from this frozen pantry.
Nutrition values are per 100 g. Quantities are exact integer grams.

Pantry:
- almonds: available=100g, step=25g, minimum-if-used=50g; energy=584kcal, protein=21.5g,
  fiber=10.8g, sodium=0.0mg; tags=tree_nut
- banana: available=125g, step=25g, minimum-if-used=50g; energy=97.0kcal, protein=0.74g,
  fiber=4.62g, sodium=0.0mg; tags=none
- carrots: available=150g, step=25g, minimum-if-used=50g; energy=45.0kcal, protein=0.941g,
  fiber=3.1g, sodium=86.6mg; tags=none
- grape_tomatoes: available=100g, step=25g, minimum-if-used=50g; energy=27.0kcal,
  protein=0.83g, fiber=2.1g, sodium=6.0mg; tags=none
- navel_orange: available=150g, step=25g, minimum-if-used=50g; energy=47.0kcal,
  protein=0.91g, fiber=2.0g, sodium=9.0mg; tags=none
- sunflower_seeds: available=125g, step=25g, minimum-if-used=50g; energy=571kcal,
  protein=18.9g, fiber=7.22g, sodium=0.0mg; tags=none

Use 2 to 4 ingredients.
Forbidden tags: none
Targets:
- mass_g: minimum 125, maximum 200
- energy_kcal: minimum 239.4, maximum 432.25
- protein_g: minimum 6.423
- fiber_g: minimum 3.478
- sodium_mg: maximum 37.6

Choose the ingredient support as a six-bit mask in the exact Pantry row order. Bit 1
includes that row and bit 0 excludes it.<|im_end|>
<|im_start|>assistant

```

F DATASET AND IDENTITY AUDITS

Main-body link. This appendix audits the materialization and identity claims in Section 3 and Table 3.

PantryPlan audit. The deterministic v2 release contains 384 train, 64 development, and 128 evaluation prompts across four formulation families; exact exhaustive support counts range from

Domain	Generated answer / check	Executed key	Aliases that merge	valid modes
Graph coloring	Complete assignment; check fixed colors and edges	Color vector	Spacing only	6.27
Countdown	Parse operands; execute exactly	Normalized AST	Commutative order	4.52
Python factors	Execute restricted lambda externally	Return vector	Source programs with same behavior	229.44
MathIR	Execute bounded action program	State trajectory	Programs with the same states	5.00
PantryPlan	Exact ingredient allocation; check all bounds	Ingredient support	Quantity and ordering aliases	18.19
PointMaze	Select adjacent move; simulate to terminal success	Directed corridor	Trajectories crossing the same corridor	3–5

Table 3: **MODEBENCH**. Correctness and identity come from one executable path; training starts without the mode catalogue. *Valid modes* is the mean number of correct execution modes per evaluation prompt, exact over the released multi-answer split; for PointMaze it is the number of separated corridors a map exposes. Every domain trains on 384 prompts and evaluates on a fixed 128-prompt split; the exact prompt each policy receives is reproduced verbatim in Appendix E.

	Python factors	MathIR action menu
Training rows	384	384
Evaluation rows	128	128
Exact modes per prompt	16–3,600	5
Training-row SHA-256	bcfa9acc...37cf7	bfc98362...5d3835
Evaluation-row SHA-256	be0f621c...94d183	df5783dd...328b96
Gold catalogue supplied to policy	no	no
Training-bank initialization	empty	empty

Table 4: Deterministic materialization identities. For every Python row, two independently executed functions must verify with different keys. Every MathIR certified route is executed by the same interpreter used for model outputs.

8 to 45. The reported comparison uses the full 384-prompt training split and the fixed 128-row evaluation split, whose 2,328 feasible supports average 18.1875 per prompt. The train/evaluation row hashes are a4883c5e...a757f and 5d81ba8d...5923. The terminal audit status is `pass` with decision `pantry_terminal_eligible`. Correctness checks exact inventory, serving steps, mass, nutrient bounds, and dietary tags; identity is ingredient support. The benchmark does not claim that a verified allocation is a recipe, palatable, cookable, or safe.

PointMaze prospective audit. The frozen generator seed is 88,104 and the frozen split sizes are 384 train, 64 development, and 128 evaluation maps. Before any training cell is eligible, the preparation audit must verify balanced rotations, three to five separated corridor identities per map, disjoint fingerprints from every interface-development cohort, and successful execution of every advertised corridor in the pinned MuJoCo runtime. Those certified route programs are audit witnesses only: they never enter the policy prompt, warm start, online training, or replay bank. There is no outcome-dependent viability gate and no replacement of a scientific cell.

G ALGORITHMIC DETAILS

Main-body link. This appendix formalizes the state transition summarized by Figure 3 and Section 4. For each prompt x , let $\mathcal{B}_x$ be the capacity- M verified-exemplar bank and let τ be the recurrent replay cursor.

Algorithm 1 One verified-replay Dr.GRPO optimizer update

Require: Policy π_θ ; prompt x and verifier specification s_x ; group size G ; replay weight λ_{replay} ; states $(\mathcal{B}, \tau)$
Ensure: Atomically updated $(\theta', \mathcal{B}', \tau')$

- 1: Draw the on-policy group $\mathcal{G} = \{y_i \sim \pi_\theta(\cdot | x)\}_{i=1}^G$
- 2: **for** $i = 1, \dots, G$ **do**
- 3: Execute $\tilde{c}_i \leftarrow V(y_i, s_x)$ and observe task reward r_i
- 4: Set $c_i \leftarrow \tilde{c}_i$ iff $r_i > 0 \wedge \tilde{c}_i \neq \perp$; otherwise set $c_i \leftarrow \perp$
- 5: **end for**
- 6: Compute centered Dr.GRPO task advantages A_i^{task} from $\{r_j\}_{j=1}^G$
- 7: After the full group is validated, retain the first policy exemplar for each new $c_i \neq \perp$ in $\mathcal{B}_x$, up to capacity M
- 8: $(z, \tau') \leftarrow \text{NEXTBANK}(\{\mathcal{B}_u\}_u, \tau)$ by a fixed round-robin cursor
- 9: **if** a replay bank $\mathcal{B}_z$ exists **then**
- 10: For each retained exemplar e_c , compute $s_c = |e_c|^{-1} \log \pi_\theta(e_c | z)$
- 11: $L_{\text{replay}} \leftarrow -|\mathcal{B}_z|^{-1} \sum_{c \in \mathcal{B}_z} s_c$
- 12: **else**
- 13: $L_{\text{replay}} \leftarrow 0$
- 14: **end if**
- 15: $w_{\text{rep}} \leftarrow (G - 1)/G^2$ for one replay pseudo-group per G on-policy rows
- 16: $\theta' \leftarrow \text{UPDATE}(\theta, L_{\text{PPO}}(A^{\text{task}}) + w_{\text{rep}} \lambda_{\text{replay}} L_{\text{replay}})$
- 17: Checkpoint $(\theta', \mathcal{B}', \tau')$ as one atomic state

The compute-matched control executes the same bank update, scheduling, teacher-forced forward pass, and backward traversal, but multiplies the applied replay derivative by exact zero. The treatment changes only that derivative from zero to live.

H WHY X-MODE GRPO IS MINIMAL

Scope. x-Mode GRPO has three operational pieces: executable mode identity, persistent verified exemplars, and recurrent scheduling. Earlier semantic, explicit-balance, and support-proposal variants remain immutable development records but are not components of x-Mode GRPO.

Every superseded developmental revision remains in its original preregistration and audit record. Its result is historical provenance, not an estimate of x-Mode GRPO.

I HISTORICAL MOTIVATION FOR X-MODE GRPO

Scope. This frozen historical comparator asked whether rehearsal of verified responses was sufficient within a superseded multi-component treatment. Ordinary rehearsal used matched Dr.GRPO with replay; the comparator added a separate balance loss. The arms share one superseded objective contract and form a clean comparison within it. Neither arm estimates x-Mode GRPO.

Historical result under the superseded contract. Ordinary rehearsal improves the mean over matched Dr.GRPO in every domain, with every paired interval excluding zero. Against replay plus balance, rehearsal is equivalent on Graph, Countdown, and MathIR, better by +.992 modes on Python, and worse by -.469 on PantryPlan. This frozen comparison motivates verified replay as the common mechanism and balance as domain-dependent; it does not establish x-Mode GRPO. No effect is pooled across domains.

Evidence boundary. These frozen arms compare rehearsal and a separate balance loss within their superseded objective contract, but neither is an estimate of x-Mode GRPO. The clean x-Mode GRPO cohort will replace this table. The historical artifact is `var/artifacts/e72_b1b_summary.json`, generated by `ops/exp_scaling/aggregate_e72_b1b.py`; its registered protocol is `paper/preregistration/e72_b1b_rehearsal_only_replay_05b_20260801.md`.

Retained piece	Role	Evidence and allowed conclusion
Executable gate and canonical key	Admit only successful outputs and merge surface aliases before replay.	Validator contracts and identity audits (Sections 3, Appendices D–F); definitional and engineering evidence, not a policy-effect estimate.
Uniform verified replay	Give every retained policy-generated success a gradient after it disappears from the current on-policy group.	Theorem A.4 proves banked-mode non-extinction in the categorical mean flow; Corollary A.5 gives the full-coverage no-collapse result. The clean five-domain neural comparison is the running x-Mode GRPO cohort and is reported only after its frozen pass-8 endpoint completes.
Deterministic recurrent schedule	Revisit banks between prompt occurrences under a fixed one-bank compute budget.	The schedule is part of the frozen replay treatment and its compute-matched control. The clean x-Mode GRPO comparison estimates the whole replay mechanism, not a separate scheduling effect.
Exact-zero replay control	Isolate the replay derivative from bank bookkeeping and teacher-forced traversal cost.	The registered comparison changes the applied replay derivative from exact zero to live while matching model, data, seed, GPU model, optimizer, scheduler, and replay compute within every pair.

Table 5: **Evidence for the minimal x-Mode GRPO design.** Semantic shaping, a separate balance KL, and support proposals are excluded from x-Mode GRPO and from the clean comparison.

Domain	Dr.GRPO	rehearsal	replay+balance	balance effect
Graph coloring	.325	2.439	2.505	E
Countdown	.627	1.884	1.846	E
Python factors	.172	1.686	.694	rehearsal better
MathIR action menu	.666	.912	.963	E
PantryPlan	.634	1.887	2.356	balance better

Table 6: **Historical ordinary rehearsal and replay plus balance.** Five paired seeds 43–47 at terminal pass 12 under the superseded objective contract; entries are mean `distinct@8`. E denotes equivalence under the frozen paired-bootstrap rule with margin $\delta = .15$. The two directional rows have intervals $[+.733, +1.324]$ for rehearsal minus replay+balance on Python and $[-.677, -.265]$ on PantryPlan.

J DECODING CONTROL: DOES ANY SAMPLER SETTING REPAIR THE COLLAPSE?

Scope. This appendix records a sampler control for a superseded 12-pass cohort. It is not evidence for x-Mode GRPO.

Every headline cell samples at temperature one. A reader may reasonably ask whether matched Dr.GRPO’s missing modes are merely improbable rather than absent: if a modestly hotter sampler recovered them, the finding would concern sharpening at a fixed decoding setting, not loss of support. That is a different scientific claim, so we test it directly.

Protocol. We re-measure all 50 frozen terminal checkpoints (five domains $\times$ two arms $\times$ seeds 43–47) at six sampling temperatures $T \in \{0.5, 0.7, 1.0, 1.3, 1.6, 2.0\}$, giving 300 terminal-checkpoint cells plus 60 hardware-matched pre-RL controls; at a four-fold larger sample budget, $K = 32$, over $T \in \{1.0, 1.3, 1.6\}$, for a further 150 terminal-checkpoint cells plus 30 pre-RL controls; and under

nucleus truncation, top- $p = 0.95$, at $T \in \{1.0, 1.6\}$, for a final 100 terminal-checkpoint cells plus 20 pre-RL controls. The three knobs a practitioner actually turns are therefore each measured against the same frozen checkpoints, 660 cells in all. Each cell loads one checkpoint and runs the *same* evaluation path used during training—the same validator, prompts, split, response budget, registered K , and four fixed draw seeds—while performing no rollout, optimizer step, or export, so a cell is a pure function of the checkpoint and its decoding setting. Two controls make the sweep admissible. First, every cell at $T = 1$ must reproduce its run’s published terminal value within $\max(3 \text{ SE}, .02)$ for rates and $\max(3 \text{ SE}, .05)$ for mode counts; all 240 such checks pass, the great majority exactly. Second, each cell runs on the GPU model that trained its checkpoint, because floating-point reduction order differs between GPU models and changes which tokens are sampled from identical weights under identical draw seeds. That effect is visible in the frozen pass-0 rows, which are identical within each GPU group and differ between them. It also bears on the headline cohort: twenty-two of the twenty-five paired seeds ran both arms on one GPU model and three pairs are split, so the sweep below pins placement where the headline design does not.

Estimand. For arm a and domain d define the *temperature-repair index*

$$\rho_{a,d} = \frac{\max_T \text{distinct@8}(a, d, T)}{\text{distinct@8}(\text{historical multi-component treatment}, d, 1.0)}.$$

The numerator maximizes over the whole sweep *with no accuracy constraint*, so the baseline may spend arbitrary correctness to buy breadth. A value below one means that no decoding setting recovers the treatment’s temperature-one breadth.

Domain	matched Dr.GRPO			historical multi-component treatment	
	best #modes	at T	m@8 there	#modes at $T=1$	ρ
Graph coloring	.327	1.3	.322	2.406	.14
Countdown	.650	1.6	.523	1.893	.34
Python factors	.172	0.5	.172	1.594	.11
MathIR action menu	.675	2.0	.363	.926	.73
PantryPlan	.719	2.0	.522	2.186	.33

Table 7: **Temperature does not repair the baseline.** Each row reports the largest distinct@8 matched Dr.GRPO reaches at *any* of the six temperatures, the temperature that achieves it, and the accuracy it costs, against historical multi-component treatment at the reported temperature-one operating point. Every entry is a five-seed mean at the terminal pass 12 checkpoint.

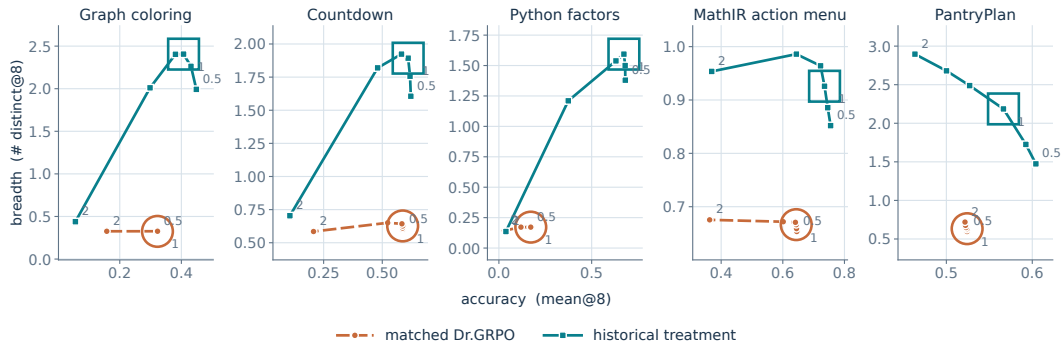


Figure 5: **Breadth against accuracy, parameterized by decoding temperature.** Each curve sweeps one arm’s terminal checkpoints over $T \in \{0.5, \dots, 2.0\}$; ringed markers are the reported temperature-one operating point and labels give T . Raising temperature moves a policy *along* its own curve, so “sample hotter” is a visible movement rather than an untested objection. The baseline curves are nearly vertical: accuracy falls while breadth stays where it was.

Result. In four of the five domains the baseline’s breadth is nearly invariant to temperature. Graph coloring moves $.323 \rightarrow .327$ across the entire range, and Python factors is constant at $.172$ through

$T = 1.3$, while accuracy degrades sharply—Python `mean@8` falls $.172 \rightarrow .029$ at $T = 2$. Heating the sampler produces invalid text, not a second verified execution. PantryPlan is the one domain where temperature does buy the baseline breadth at no accuracy cost ($.597 \rightarrow .719$ as T rises to 2 with `mean@8` steady near $.52$), which its eight-token support-mask interface makes possible because it has no malformed-output failure mode; even there the gain closes about a third of the gap. Across all five domains $\rho \leq .73$, and in four of five historical multi-component treatment’s breadth at *every* temperature exceeds the baseline’s breadth at its *own best* temperature; on Python that holds at every $T < 2$, where both arms stop returning valid output.

The same sweep shows the reported operating point is not tuned in the treatment’s favor: historical multi-component treatment’s breadth peaks at $T = 1.0$ – 1.3 in four domains, so temperature one is at or near its own optimum. PantryPlan is again the exception, with historical multi-component treatment continuing to 2.896 at $T = 2$; reporting temperature one understates the treatment there.

A larger sample budget does not repair it either. “Sample hotter” has a budget-shaped twin—“sample more”—and it deserves the same kind of answer. Holding temperature at the reported operating point and moving only K from 8 to 32 quadruples the draws while changing nothing else. Table 8 reports what each arm does with them.

Domain	matched Dr.GRPO			historical multi-component treatment		
	d@8	d@32	Δ	d@8	d@32	Δ
Graph coloring	.325	.327	+0.002	2.406	4.237	+1.832
Countdown	.627	.644	+0.017	1.893	2.239	+0.346
Python factors	.172	.172	+0.000	1.594	2.137	+0.543
MathIR action menu	.666	.680	+0.013	.926	1.076	+0.150
PantryPlan	.634	.637	+0.004	2.186	4.728	+2.542

Table 8: **Breadth against sample budget at temperature one.** `d@K` is `distinct@K`; five-seed means at the terminal pass 12 checkpoint, with decoding otherwise identical to the reported operating point. Quadrupling the budget moves the baseline by at most $.017$ modes and moves historical multi-component treatment by up to 2.542 .

The baseline does not convert budget into breadth in any domain: the largest gain across all five is $.017$ modes, and Python factors gains exactly nothing. Its `pass@K` is flat for the same reason— $.594 \rightarrow .594$ on Countdown, $.556 \rightarrow .556$ on PantryPlan, $.172 \rightarrow .172$ on Python factors—so twenty-four additional draws per prompt surface no verified execution the first eight did not already contain. That is the sharpest available statement of the failure: the missing modes are not rare, they are absent, and no amount of sampling from this policy will produce one. Historical multi-component treatment spends the same budget very differently, nearly doubling its breadth on Graph coloring and more than doubling it on PantryPlan, which is what a policy that has retained its support looks like under a wider draw.

The hardware-matched pre-RL controls make the support loss literal in the two domains where initialization already has broad verified support. At $K = 32$, the two hardware-specific references yield 1.590 – 1.723 Graph-coloring modes and 7.129 – 7.258 PantryPlan plans. Terminal matched Dr.GRPO yields only $.327$ and $.638$, respectively, while historical multi-component treatment yields 4.238 and 4.728 . Matched Dr.GRPO therefore does not merely fail to discover alternatives: it removes verified support already observable before RL. On PantryPlan, historical multi-component treatment mitigates but does not fully restore the pre-RL breadth.

Nor does truncating the tail. The third knob cuts the other way: nucleus sampling deletes low-probability mass, so if the baseline’s missing modes were merely improbable, truncation should visibly cost it breadth. At the reported operating point it costs it nothing. Moving to $\text{top-}p = 0.95$ changes matched Dr.GRPO’s `distinct@8` by $+0.000$ on Graph coloring and Python factors, -0.001 on MathIR and PantryPlan, and -0.021 on Countdown. Historical multi-component treatment, by contrast, does lose breadth to the same operation— $.126$ on PantryPlan and $-.020$ on Graph coloring—because it has tail mass to lose. A policy whose support has collapsed is indifferent to how the tail is cut, and that indifference is itself the measurement.

The practitioner’s usual recipe for diverse-but-valid sampling is to raise temperature and truncate together. At $T = 1.6$ with top- $p = 0.95$ that recipe recovers real breadth for the treatment, +.472 on Python factors and +.256 on Graph coloring over untruncated sampling at the same temperature, because truncation removes the malformed text that heating produced. For the baseline the same combination returns +.000 on Graph coloring, Python factors, and MathIR. There is nothing in the tail to keep.

We therefore read the failure as loss of support rather than sharpening: the absent modes are not waiting at low probability for a hotter sampler, a wider one, or a narrower one. This control covers decoding of the trained policies only. It says nothing about the alternative training objectives listed in Section 8.

K CROSS-FAMILY REPLICATION ON A SECOND BASE MODEL

Scope. This appendix records a historical second-base-model replication of a superseded multi-component method. It is not evidence for x-Mode GRPO.

Why. Every headline cell is produced by one base model. A reader may reasonably ask whether the collapse of matched Dr.GRPO’s metric band, and its reopening under historical multi-component treatment, is a property of the objective or of that one model family’s tokenizer and instruction tuning. The registered replication answers that directly by re-running the whole five-domain design on Falcon3-1B-Instruct at a pinned revision: an independent line in tokenizer, pretraining corpus, and instruction tuning, and roughly three times the parameter count. The parameter count is deliberately not matched, because no comparison in the study is between the two base models; every comparison is within a domain, between the two arms, at a matched seed.

What is held fixed. The five domains, both arms, seeds 43–47, 12 epochs over 384 training prompts with a fixed 128-prompt evaluation split, group size 16, $\beta_{KL} = 0$, rollout temperature 1, the evaluation protocol, the reporting rule, and every historical multi-component treatment coefficient at its value in this paper. No coefficient is retuned for the new model.

What necessarily changes. Three things, each forced by the base model rather than chosen, and each applied identically to both arms. First, the *chat surface*: Falcon3 was instruction-tuned on different role markers, so each domain uses a twin of its prompt contract that shares the system instruction and the canonical answer rewrite verbatim and differs only in those markers; the rendering reproduces the model’s own published chat template byte for byte. Second, *optimizer-state placement*: at 1.67B parameters the weights, master copy, and Adam moments do not fit a 24 GB device alongside a colocated inference engine, so optimizer state is offloaded to host memory, which changes where state lives and not the update computed. Third, two *response budgets*. Falcon3 is uniformly more verbose than Qwen2.5-0.5B, and budgets calibrated to the latter’s terseness truncate it: measured against the paired runs before any outcome was read, the manuscript’s budget bound 44% of Falcon rollouts on Python factors and 14% on MathIR while binding the Qwen cohort on 20 of 4,609 and 0 of 4,609 responses respectively. Those two budgets were raised until truncation was negligible. Graph coloring (6%) and Countdown (2%) were left at the manuscript values, and PantryPlan is unaffected because its canonical sampler emits a fixed six-token action.

What transfers without modification. The canonical action spaces resolve to the same horizon, sequence count, and maximum sequence entropy under the new tokenizer as under the original: 3/27 for Graph coloring, 3/108 for Countdown, and 6/64 for PantryPlan. The objective is therefore defined over the same action geometry in both families, and the reward and validator paths are surface-independent.

Registered prediction. Stated before launch and deliberately qualitative: matched Dr.GRPO’s four metrics will again coincide, with modes-per-success near one, in at least four of the five domains; historical multi-component treatment will raise `distinct@8` in at least four of the five; and historical multi-component treatment will not reduce `pass@8` below the matched baseline in any domain. Absolute values are *not* predicted to reproduce this paper’s numbers, and no threshold on their agreement is registered. A different base model has different task competence; what is under test is the direction and the presence of the band collapse, not the magnitude.

Claim boundary. A negative or mixed result is reportable and will be reported. If the effect does not reproduce, the correct conclusion is that the result as stated is contingent on the base model, and this paper’s claim will be narrowed accordingly rather than the cohort excluded. Because two response budgets differ from the manuscript’s, the replication’s Python factors and MathIR rows are not comparable to a run of the same model at the original budgets, and the reported cohort uses the raised-budget runs throughout.

Domain	Arm	p@1	mean@8	pass@8	#modes	modes/succ.
Graph coloring	matched Dr.GRPO	.277	.281	.546	0.78	1.42
	historical multi-component treatment	.377	.358	.822	1.70	2.07
Countdown	matched Dr.GRPO	.314	.315	.368	0.39	1.06
	historical multi-component treatment	.417	.378	.668	0.86	1.29
Python factors	matched Dr.GRPO	.000	.000	.000	0.00	–
	historical multi-component treatment	.897	.789	.964	1.26	1.30
MathIR	matched Dr.GRPO	.070	.070	.070	0.07	1.00
	historical multi-component treatment	.653	.508	.862	0.92	1.06
PantryPlan	matched Dr.GRPO	.633	.633	.633	0.63	1.00
	historical multi-component treatment	.577	.551	.778	1.72	2.21

Table 9: **Terminal five-domain cross-family replication.** Falcon3-1B-Instruct, five-seed (43–47) means of both arms at the terminal epoch 12, using the preregistered deepest-common-checkpoint rule. **#modes** is `distinct@8`, a count of distinct verified modes per eight-sample draw. Historical multi-component treatment leads on `distinct@8` and `pass@8` in all twenty-five paired seed cells. A dash marks Python’s undefined baseline modes-per-success because its terminal `pass@8` is zero, not a carried-forward value.

Terminal result. The two method-direction clauses registered before launch are met with room to spare. Historical multi-component treatment raises `distinct@8` by +.923, +.475, +1.255, +.846, and +1.084 modes on Graph, Countdown, Python, MathIR, and PantryPlan, and raises `pass@8` by +.276, +.300, +.964, +.792, and +.145. Both differences are positive in every paired seed: 25 of 25 for breadth and 25 of 25 for success. PantryPlan reproduces the central accuracy–support dissociation particularly sharply: baseline greedy accuracy is higher (.633 versus .577), yet historical multi-component treatment returns 1.72 versus 0.63 modes and raises `pass@8` from .633 to .778.

The baseline-collapse clause needs a narrower verdict. Countdown, MathIR, and PantryPlan end at modes-per-success 1.06, 1.00, and 1.00; Graph is the exception at 1.42. Python’s four baseline metrics all equal zero, so its modes-per-success is undefined. The protocol specified neither a numerical tolerance for “near one” nor a convention for zero-success endpoints. Thus the four-metric band closes qualitatively in four domains, but a strict four-of-five verdict on the compound ratio clause is not licensed. We report the two directional clauses as confirmed and the baseline-collapse clause as partially unresolved, rather than choosing a post-hoc zero convention.

Per-seed rows and paired win counts are bound in `var/artifacts/e73_falcon_cross_family_summary_v2.json`, which reads the distinct-mode count `sampld_distinct_correct_at_8` and aborts if any cell violates Lemma 2.2. Absolute agreement with Qwen was not predicted, and these fixed-recipe results are robustness evidence rather than a tuned estimate for Falcon.

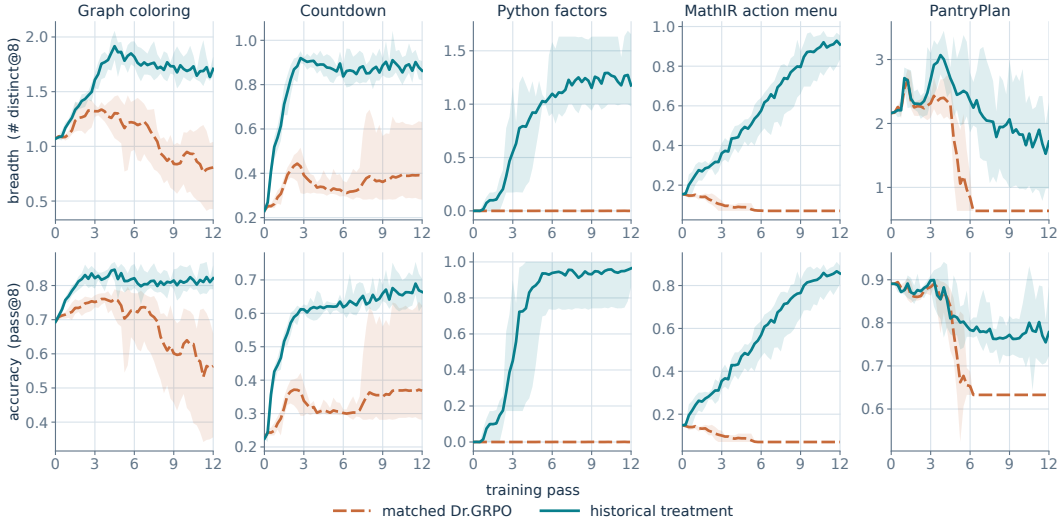


Figure 6: **Cross-family trajectories on Falcon3-1B-Instruct.** Breadth (top) and accuracy (bottom) against training pass, one column per MODEBENCH domain, five-seed means with the seed range as a band. Every panel reaches the common terminal pass 12; nothing is extrapolated and no seed is carried forward. Graph coloring shows the manuscript’s central phenomenon occurring on a second model family: the baseline’s breadth rises to roughly 1.2 modes by pass four and then *declines* to 0.78, while its accuracy stays flat—support is lost after being found, not never acquired.

What the trajectories add. Two things the endpoint table cannot show. First, the baseline’s breadth is not merely lower, it is *non-monotone*: on Graph coloring it peaks near 1.2 modes at pass four and falls by a third thereafter, and on MathIR and PantryPlan it turns down before the terminal endpoint. A method that had simply never found the modes would produce a flat line, not a hump. Second, the two arms separate early and stay separated: there is no depth at which the baseline is catching up, so the gap at pass twelve is not an artifact of stopping there.

Python factors is the limiting rescue: the baseline remains at zero while Historical multi-component treatment reaches .897 greedy accuracy, .964 pass@8, and 1.26 modes. MathIR supplies the other limiting trajectory, with all baseline metrics falling to .070 as the treatment continues upward. These are terminal shapes, not an early-checkpoint selection.

L REPRODUCIBILITY AND CLAIM BOUNDARIES

Main-body link. This appendix enforces the evidence boundaries stated in Section 6 and Section 8.

Remaining hyperparameters. The five-domain clean x-Mode GRPO comparison uses learning rate 2×10^{-7} , one PPO epoch, top- $p = 1$, eight passes (3,072 updates), half-pass evaluation and resumable checkpoints, replay weight $\lambda_{\text{replay}} = .10$, one bank selected by the fixed recurrent schedule per update, and replay capacity $M = 16$. No other auxiliary derivative is active. PantryPlan uses the same optimizer, dose, seed set, and pass grid with an eight-token response and a six-decision support mask. PointMaze keeps the same learning rate, dose, bank capacity, seeds, and eight-pass grid, with the same 3,072 updates and 192-update half-pass checkpoints as the static domains, a maximum 64-decision episode, and the shared frozen weak warm start.

Evidence and audit ledger. Every claim in this paper is listed below with the comparison that supports it and the strongest statement that comparison licenses. Scope follows completion and prospective control, not whether a result is favourable: superseded, repaired, and failed cohorts enter no estimate here.

Bank state, scheduler cursor, optimizer state, data cursor, and request-stream state are checkpointed together. Non-finite values, state mismatch, algorithm exceptions, missing terminal markers, proposal leakage into PPO, or proposal overlap with on-policy counts are fail-closed violations.

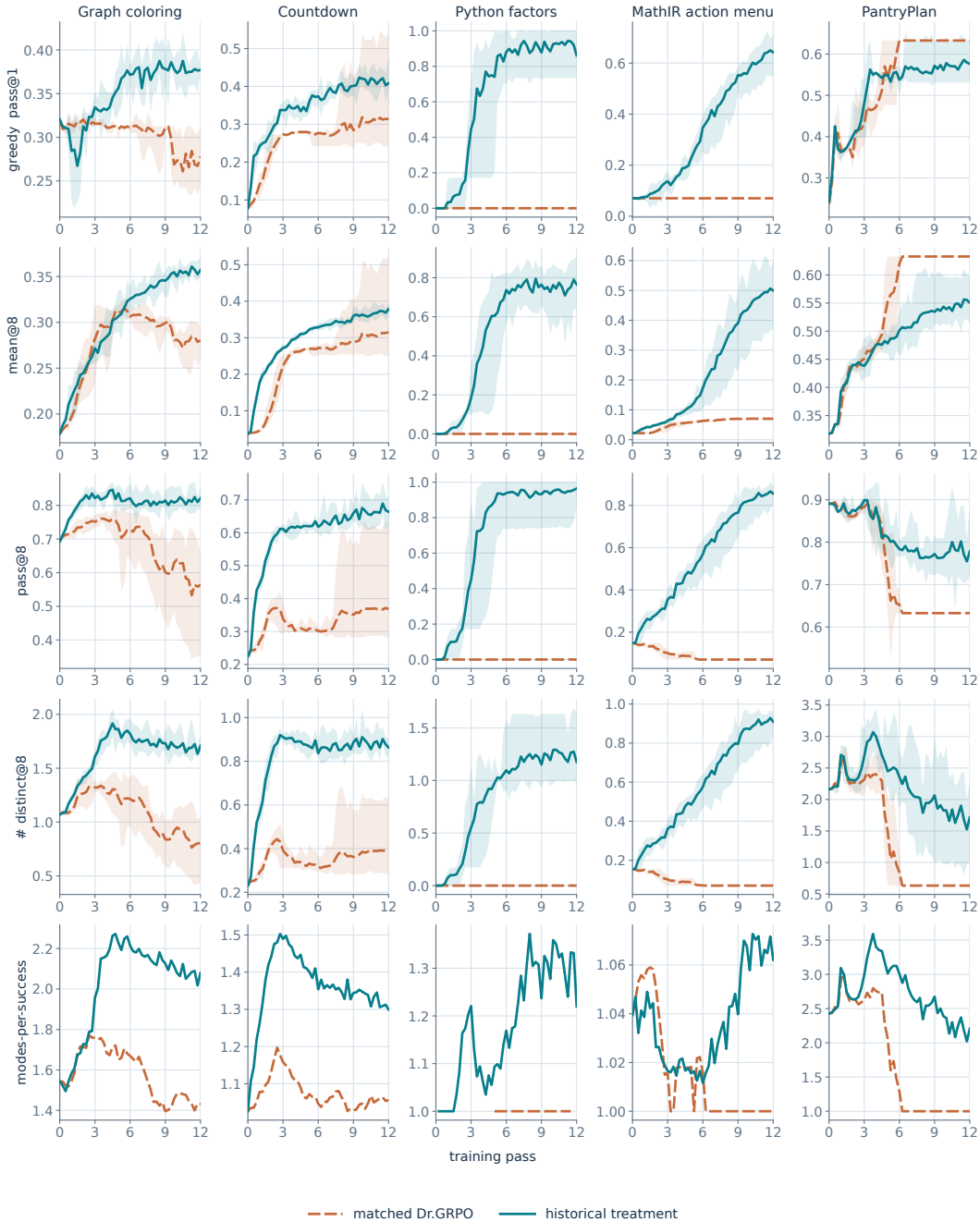


Figure 7: The whole metric band on the second family. The four reported metrics and their ratio, against training pass, one column per domain; five-seed means with the seed range as a band through terminal pass 12. The bottom row is the claim in one line: modes-per-success is $\text{distinct@8}/\text{pass@8}$, and a value of one means every successful eight-sample draw returned the same execution. Matched Dr.GRPO reaches one on MathIR and PantryPlan and approaches it on Countdown; Python ends with zero successes, so its ratio is undefined. historical multi-component treatment remains above one wherever defined. Graph coloring shows the non-monotone form: the baseline rises to 1.8 by pass four and falls to 1.42.

Placement-only recovery amendments are hash-bound. Two superseded cohorts are excluded by machine-readable objective audits rather than by their outcomes. Figure 1 is generated by `ops/plot_paper_collapse_toy.py` directly from the retained seed-43 Qwen2.5-0.5B-

Question	Comparison	Status	Allowed claim
Does x-Mode GRPO retain modes?	Five domains at 384/128, exact-zero replay derivative versus live uniform verified likelihood, seeds 43–47, eight passes	interim; running	dated descriptive trajectory only; final paired pass-8 effect remains withheld and checkpoints are not selection candidates
Does x-Mode GRPO transfer to interactive PointMaze?	Fresh 384/128 maps, exact-zero replay derivative versus live uniform verified episode likelihood, seeds 43–47, eight passes	interim; running	dated descriptive trajectory only; separate final result after all ten pass-8 endpoints, with no pooling into the static domains
Does x-Mode GRPO transfer across model scale?	Falcon3-1B seeds 55–59 and Qwen2.5-3B seeds 70–74, five static domains, exact-zero replay derivative versus live uniform verified likelihood	interim; running	currently paired trajectories only; no cross-model comparison, pooling, or final effect claim
Does the historical multi-component repair oppose collapse across the benchmark?	Five domains at 384/128, 2 arms, seeds 43–47, terminal pass 12	terminal	paired per-domain direction and magnitude on the common reporting rule; no pooled effect
Is verified replay the common retention mechanism?	Ordinary verified rehearsal and replay plus balance versus matched Dr.GRPO, 5 domains, seeds 43–47, terminal pass 12	terminal	rehearsal improves the mean in all five domains; adding balance is equivalent in three, helpful in one, and harmful in one; no pooled effect
Can any sampler setting repair the baseline?	All 50 terminal checkpoints plus hardware-matched pre-RL references re-measured over six temperatures, a four-fold sample budget, and nucleus truncation: 660 cells	terminal	decoding-level control on frozen endpoints; says nothing about alternative training objectives
Does the effect survive a change of base model?	Falcon3-1B-Instruct, same design, 2 arms, seeds 43–47; all five domains at terminal pass 12	terminal	distinct@8 and pass@8 improve in all 25 pairs; the zero-success Python endpoint leaves the compound band-collapse clause partially unresolved

Table 10: **Evidence and audit ledger.** Each row names a question, the comparison that addresses it, and the strongest claim that comparison permits.

Instruct sample logs. Its prompt-selection rule, displayed counts, source paths, and source hashes are frozen in `var/artifacts/paper_graph_collapse_toy.json`. Figure 2 is generated by `ops/plot_paper_modebench_examples.py`; its five static instance IDs are bound to the released evaluation splits under `var/data/`, and its PointMaze interface example is bound to a simulator-certified development identity. The renderer rechecks every displayed coloring edge, factor, arithmetic result, algebra state transition, maze wall, route program, and canonical corridor witness before writing the figure. Figure 3 is the component schematic generated by `ops/plot_paper_verified_replay_mechanism.py`; it contains no empirical values. Figure 4 is frozen from `ops/exp_scaling/plot_figure4_with_falcon_preview.py`. Its dated source of truth is `paper/figures/figure4_interim_20260806.json`, generated at 2026-08-06T22:46:40Z; it records observed and registered run counts, the available seeds in each arm, paired seed identities at every half pass, paired arm means and ranges, and replay-minus-control differences. Table 1 is generated by `ops/exp_scaling/build_figure4_interim_table.py`; its source of truth is `paper/results/figure4_interim_20260806_table.json`. The table fixes the same deepest pass and paired seeds as the figure snapshot, then reports their greedy pass@1 and fixed-draw sampled pass@8, mean@8, and distinct@8 means to three dec-

1728 imals. Table 6 is reproduced from `var/artifacts/e72_b1b_summary.json`, generated by
 1729 `ops/exp_scaling/aggregate_e72_b1b.py`. Table 7 and Figure 5 are produced by `ops/`
 1730 `exp_scaling/build_e72_frontier_manifest.py`, `ops/exp_scaling/launch_`
 1731 `e72_decoding_frontier.py`, `ops/exp_scaling/aggregate_e72_frontier.py`,
 1732 and `ops/exp_scaling/plot_e72_decoding_frontier.py` from `var/artifacts/`
 1733 `e72_decoding_frontier_summary.json`, with one row per measured cell in `var/`
 1734 `artifacts/e72_decoding_frontier_cells.jsonl` and the checkpoint inventory
 1735 in `var/artifacts/e72_frontier_source_runs.json`. Each cell inherits its do-
 1736 main’s evaluation configuration from the source run’s own recorded arguments, and
 1737 rendering aborts unless every temperature-one cell reproduces its published terminal
 1738 value. Table 9 is produced by `ops/exp_scaling/aggregate_e73_cross_family.`
 1739 `py` from `var/artifacts/e73_falcon_cross_family_summary_v2.json`; it reads
 1740 `sampled_distinct_correct_at_8`, the count of distinct verified modes, and aborts
 1741 if any emitted cell violates the ordering of Lemma 2.2. The running five-domain
 1742 x-Mode GRPO protocol is `paper/preregistration/e78_verified_replay_only_`
 1743 `05b_20260804.md`; PointMaze is registered separately as `paper/preregistration/`
 1744 `e78pm_point_maze_verified_replay_only_05b_20260804.md`. The aligned
 1745 Falcon3-1B and Qwen2.5-3B protocols are `paper/preregistration/e79_falcon1b_`
 1746 `aligned_verified_replay_20260804.md` and `paper/preregistration/e80r1_`
 1747 `qwen3b_aligned_verified_replay_20260805.md`; the Falcon PointMaze cohort
 1748 is registered as `paper/preregistration/e79pm_falcon_point_maze_verified_`
 1749 `replay_only_20260805.md`. All remain running and retain their original terminal es-
 1750 timands. The historical headline protocols are `paper/preregistration/e70_clean_`
 1751 `verified_maxent_vs_compute_matched_drgrpо_stage_a_05b.md` and `paper/`
 1752 `preregistration/pantry_stage_b_05b_12pass_20260730.md`; the superseded ear-
 1753 `lier protocol remains paper/preregistration/e61rl_e58_vs_grpo_05b_12pass.`
 1754 `md`. The historical base objective is bound by `paper/preregistration/e58_global_`
 1755 `verified_replay_canonical_05b.md`.
 1756 The static environments are implemented in `src/oat_drgrpо/canonical_actions.`
 1757 `py`, `src/oat_drgrpо/python_modebench.py`, `src/oat_drgrpо/mathir.py`, and
 1758 `src/oat_drgrpо/pantry_plan.py`. The external Python process boundary is in `src/`
 1759 `oat_drgrpо/python_modebench_process.py` and its worker in `src/oat_drgrpо/`
 1760 `python_modebench_worker.py`. The PointMaze environment, public decision surface,
 1761 and process boundary are in `src/oat_drgrpо/point_maze_waypoint.py`, `src/oat_`
 1762 `drgrpо/point_maze_waypoint_policy.py`, and `src/oat_drgrpо/point_maze_`
 1763 `waypoint_process.py`; its comparison runner is `ops/train_point_maze_verified_`
 1764 `replay_only.py`. Static replay is implemented in `src/oat_drgrpо/canonical_`
 1765 `replay.py`, the static verified bank in `src/oat_drgrpо/online_canonical_bank.py`,
 1766 and interactive replay in `src/oat_drgrpо/interactive_episode_replay.py`.
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